Software Quality Engineering

Lab # 08

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Submitted By:

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53143

BSSE 6-1

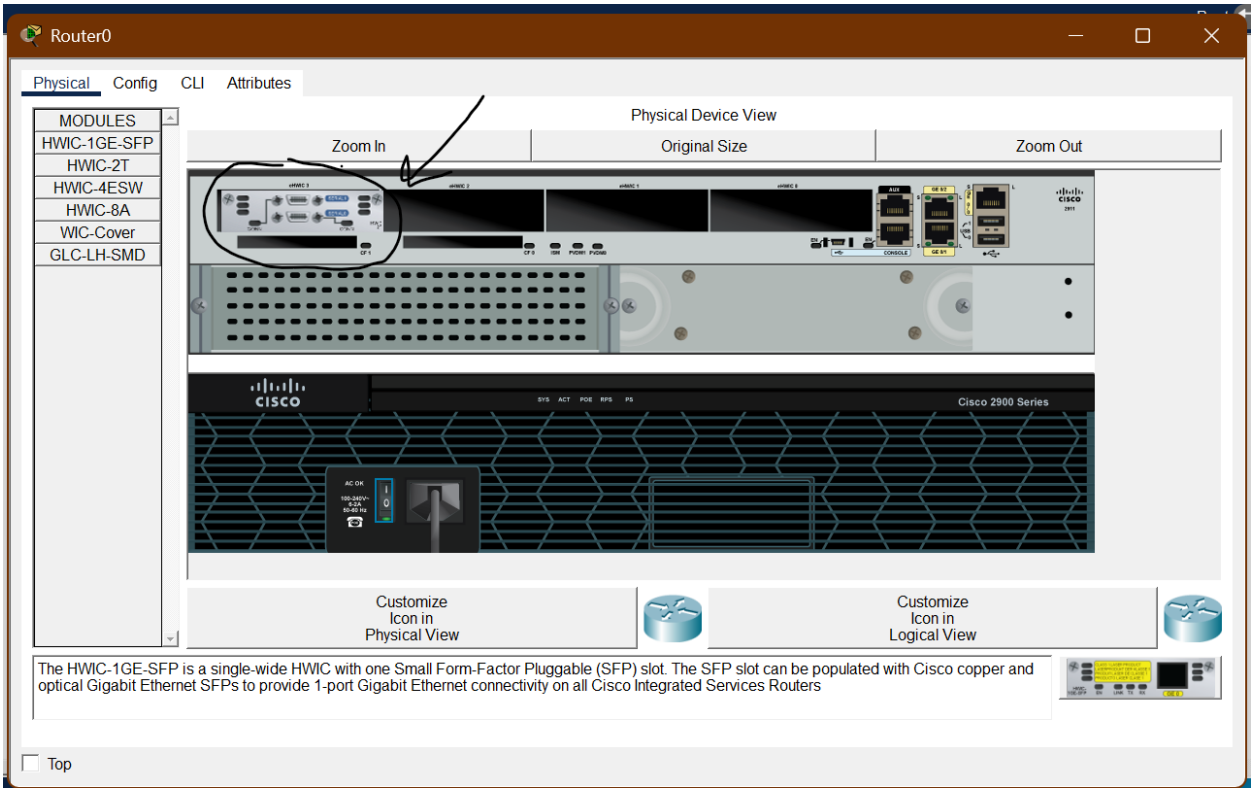
Lab Task:

Solution

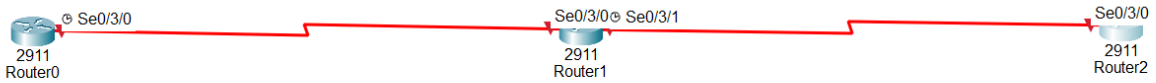
Step 1: Building the Physical Topology

1. **Routers**: Place **3 Routers** 2911 in a row:

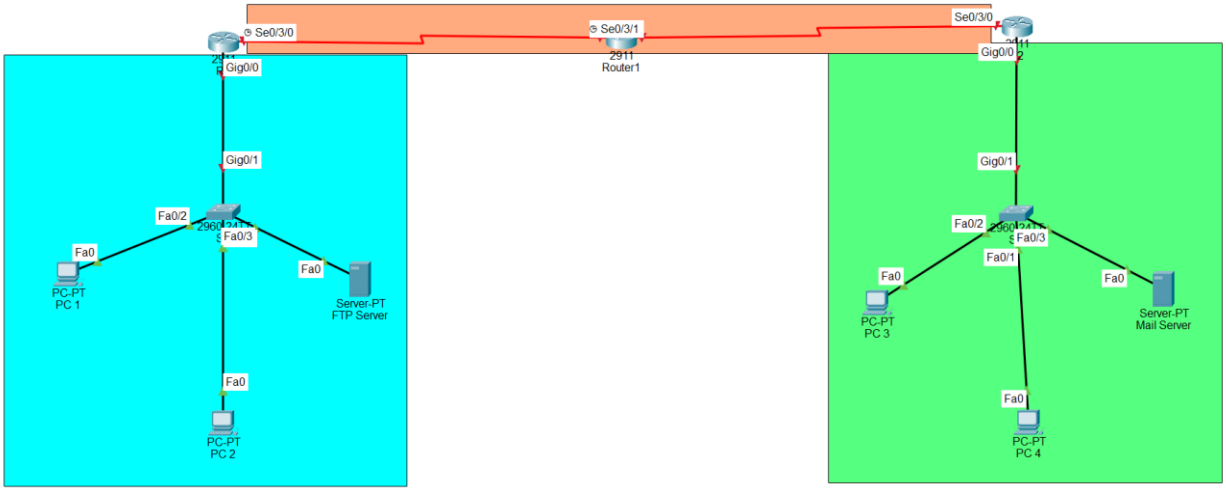
2. **Now power them off and add a HWIC-2T module to get the Serial ports needed for the WAN links.**



3. **Then connect them with "Serial DEC"cable:**

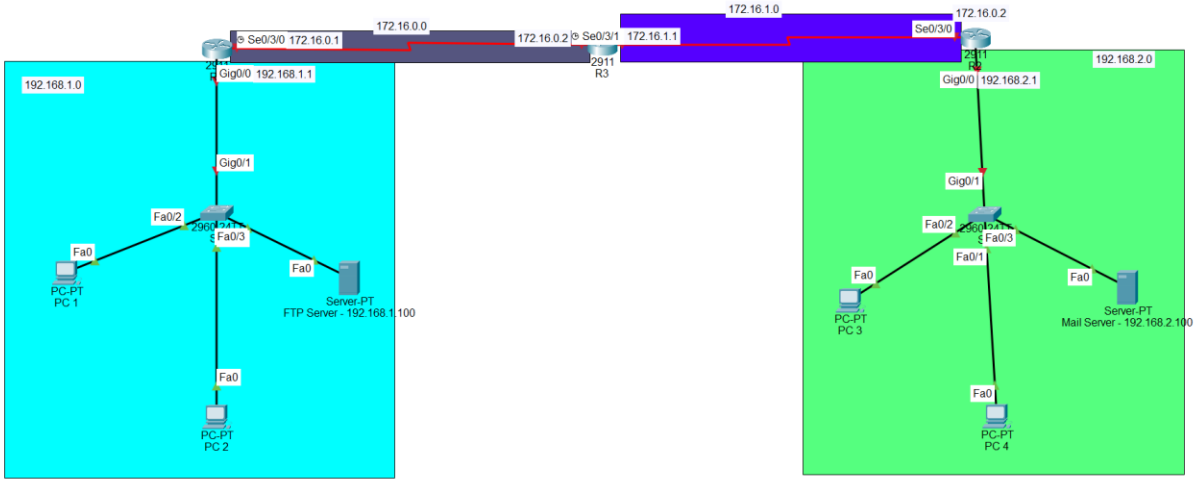


4. **Switches: Place 2 Switches (2960 models). Connect one to R1 and one to R3.**
5. **Servers: Place an FTP Server on the R1 switch and a Mail Server on the R3 switch.**
6. **End Devices: Add 2 PCs to each switch.**
7. **And connect all the devices.**



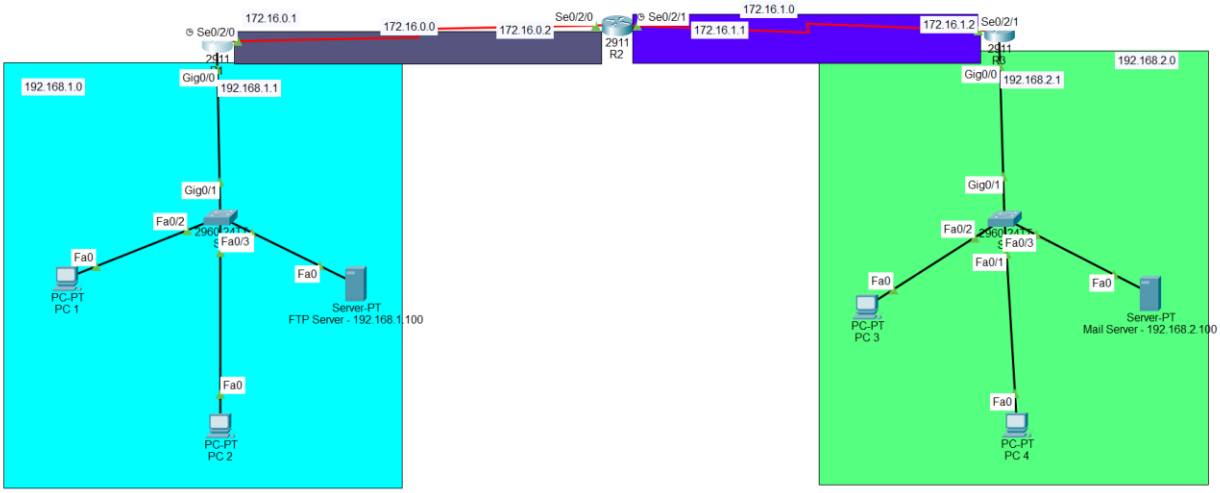
Step 2: Logical Addressing & Basic Config

1. **Assign IP addresses to every interface.**
    a. LAN 1 (R1 side): 192.168.1.0/24
    b. WAN 1 (R1-R2): 10.1.1.0/30
    c. WAN 2 (R2-R3): 10.2.2.0/30
    d. LAN 2 (R3 side): 192.168.2.0/24



Task 1: Router & Network Configuration:

1. Assign hostnames, IP addresses, and enable all required interfaces on R1, R2, and R3 according to your addressing plan



IPs assigned to each interface of router, and connection is established.

2. Secure console and VTY lines on all routers with a password and configure a login banner.
    a. Use these commands to secure console
        i. Router(config)# hostname R1
        ii. R1(config)# line console 0
        iii. R1(config-line)# password [your_password]
        iv. R1(config-line)# login
        v. R1(config)# banner motd "Authorized Access Only!"
    b. Use these commands to secure UTV lines.
        i. R1(config)# line vty 0 5
        ii. R1(config-line)# password [your_password]
        iii. R1(config-line)# login
        iv. R1(config-line)# exit
    c. Both are enabled: checked “show running-confg”

```
!
line con 0
 password 125125
 login
!
line aux 0
!
line vty 0 4
 password 125125
 login
!
```

3. Enable RIP version 2 on all routers, advertise all directly connected networks, and disable auto-summary.
    a. Use these commands to do this.
        - R1(config)# router rip

- R1(config-router)# version 2
- R1(config-router)# no auto-summary
- R1(config-router)# network [Enter your connected network IPs here]

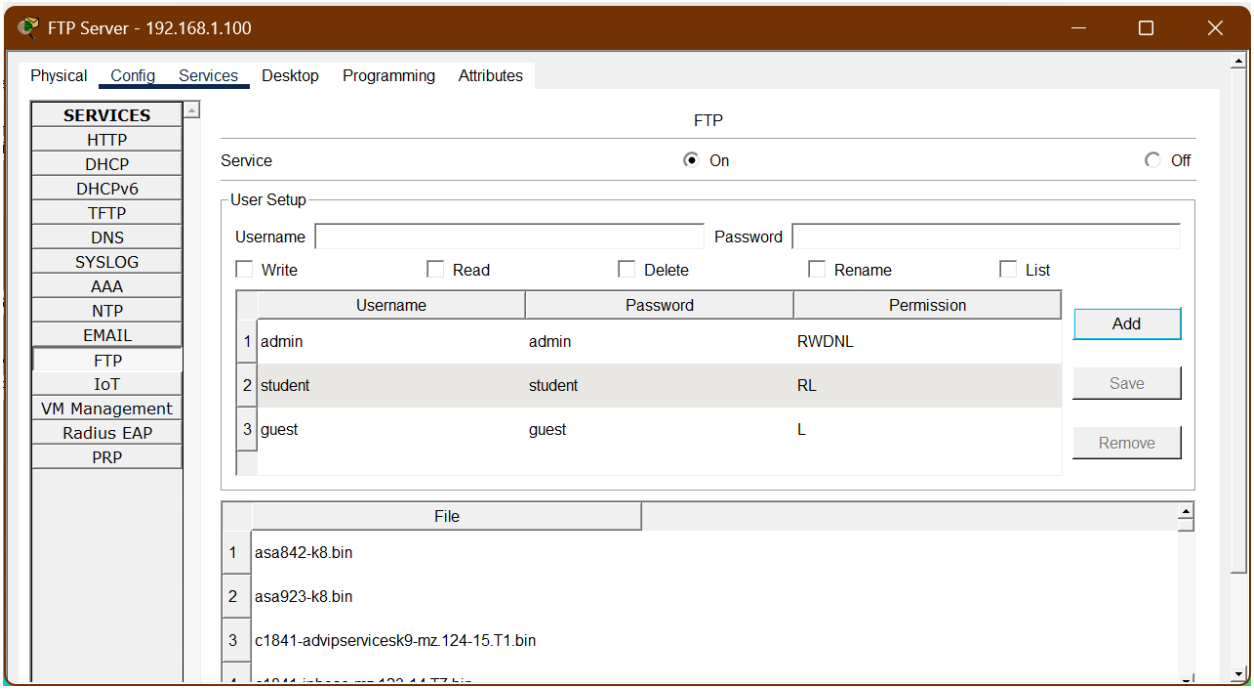
Checked using command : “show running-confg”

```
router rip
 version 2
 network 172.16.0.0
 no auto-summary
!
```

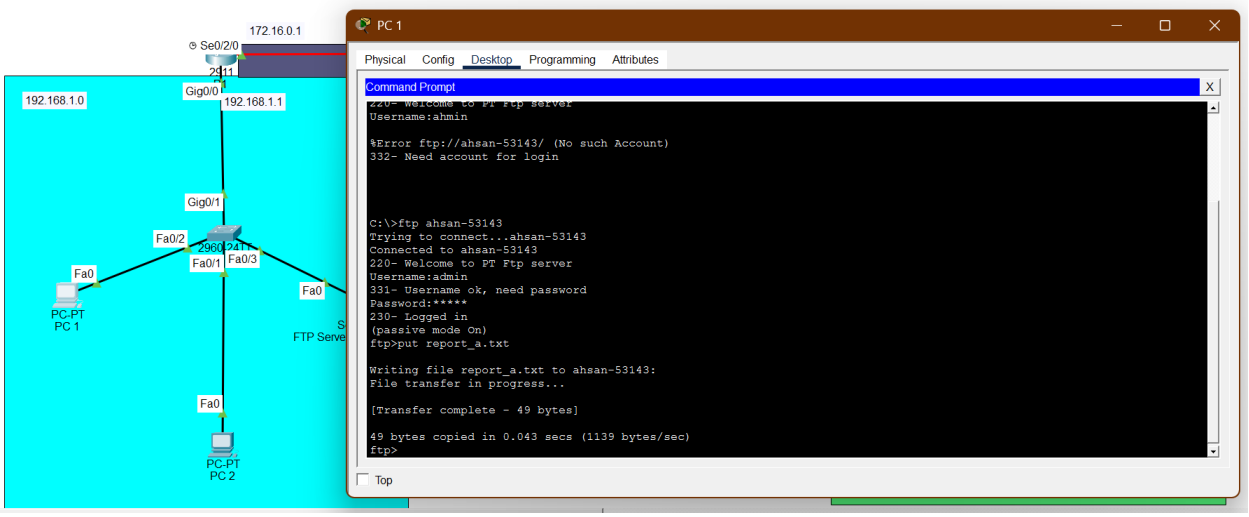
4. Configure R2 with a passive interface on one of its serial links and document the effect this has on routing updates and convergence. Restore the configuration afterward.
   - Commands to enable:
     - R2(config)# router rip
     - R2(config-router)# passive-interface Serial0/3/0

Task 2 — FTP Server Configuration & Access Control

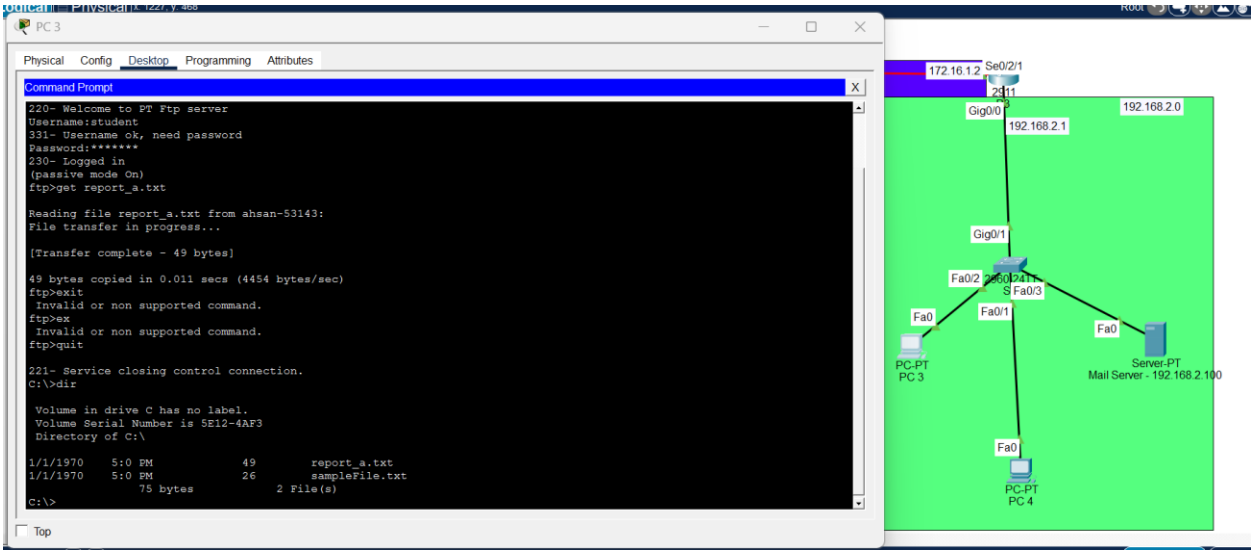
1. Enable the FTP service on the FTP Server.
2. Create three user accounts with the following access levels:
   a. admin — full permissions (read, write, delete, list, rename)
   b. student — read and list only
   c. guest — list only



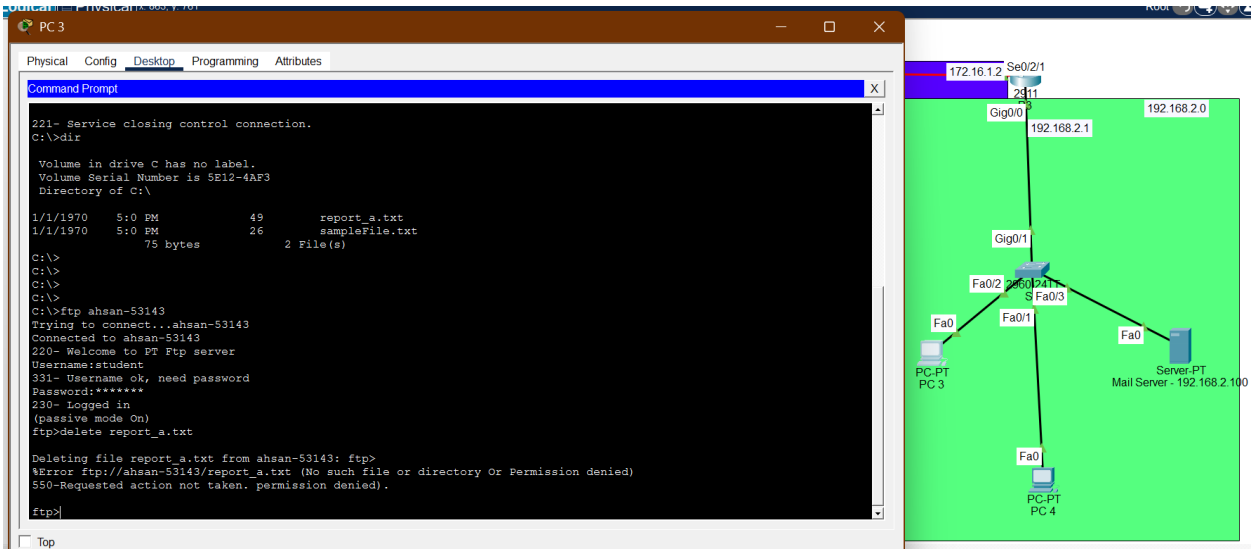
3. Create three user accounts with the following access levels:
4. From PC-A, connect to the FTP server as admin and upload a file named report_a.txt .



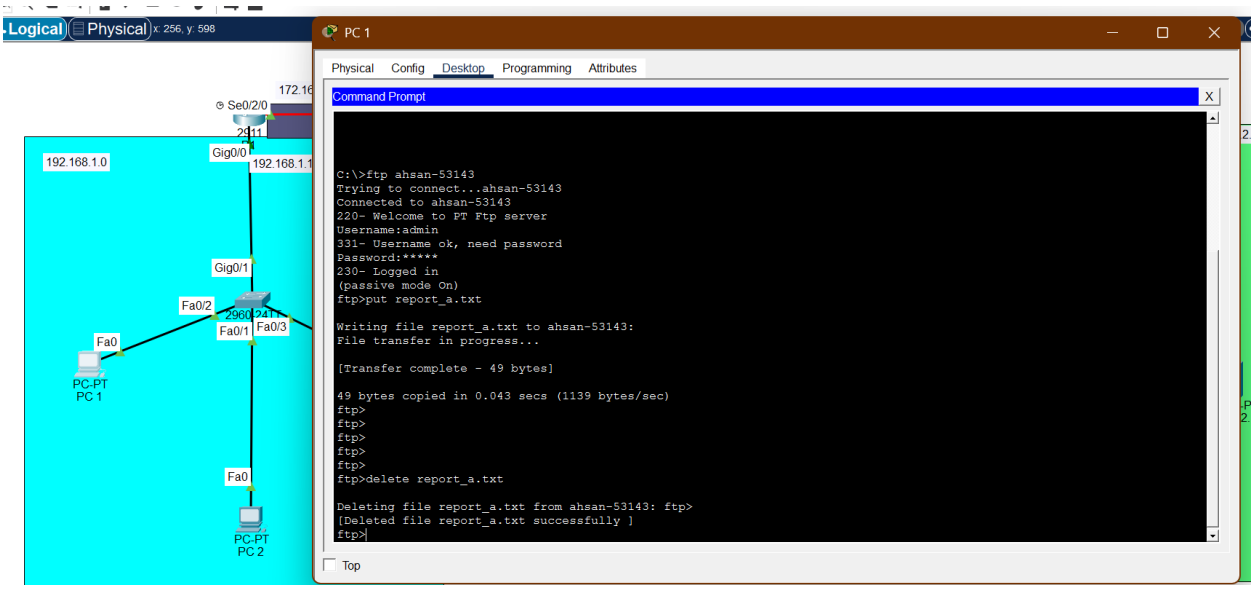
5. From PC-C (across the routed network), log in as student and download report_a.txt .



6. While still logged in as student, attempt to delete report_a.txt and record the server's response.



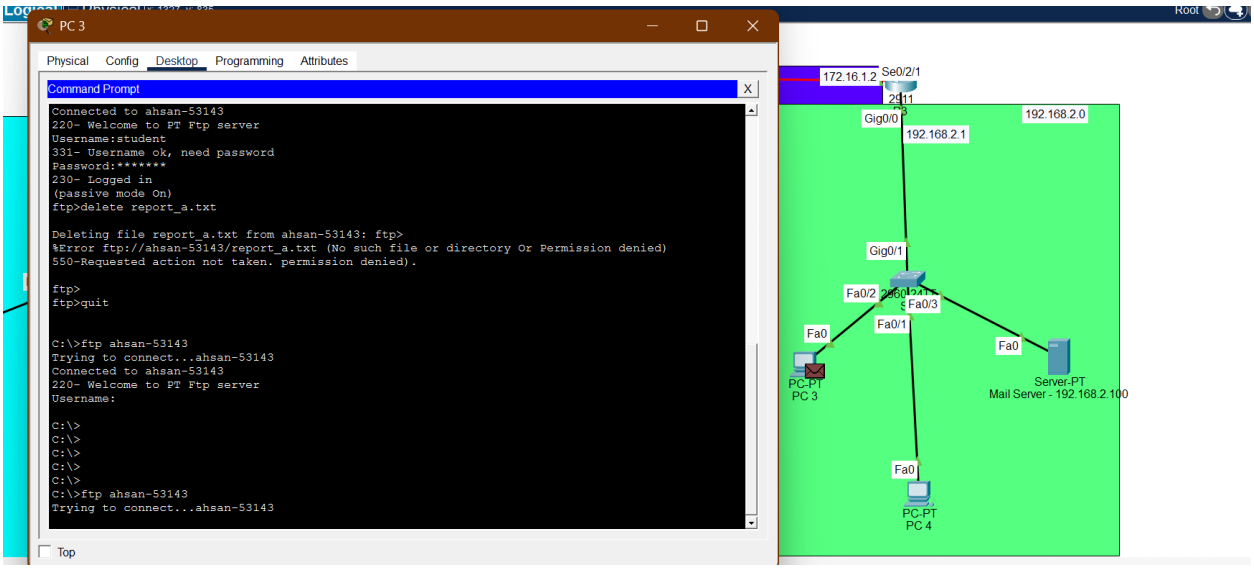
7. Log back in as admin and successfully delete report_a.txt .



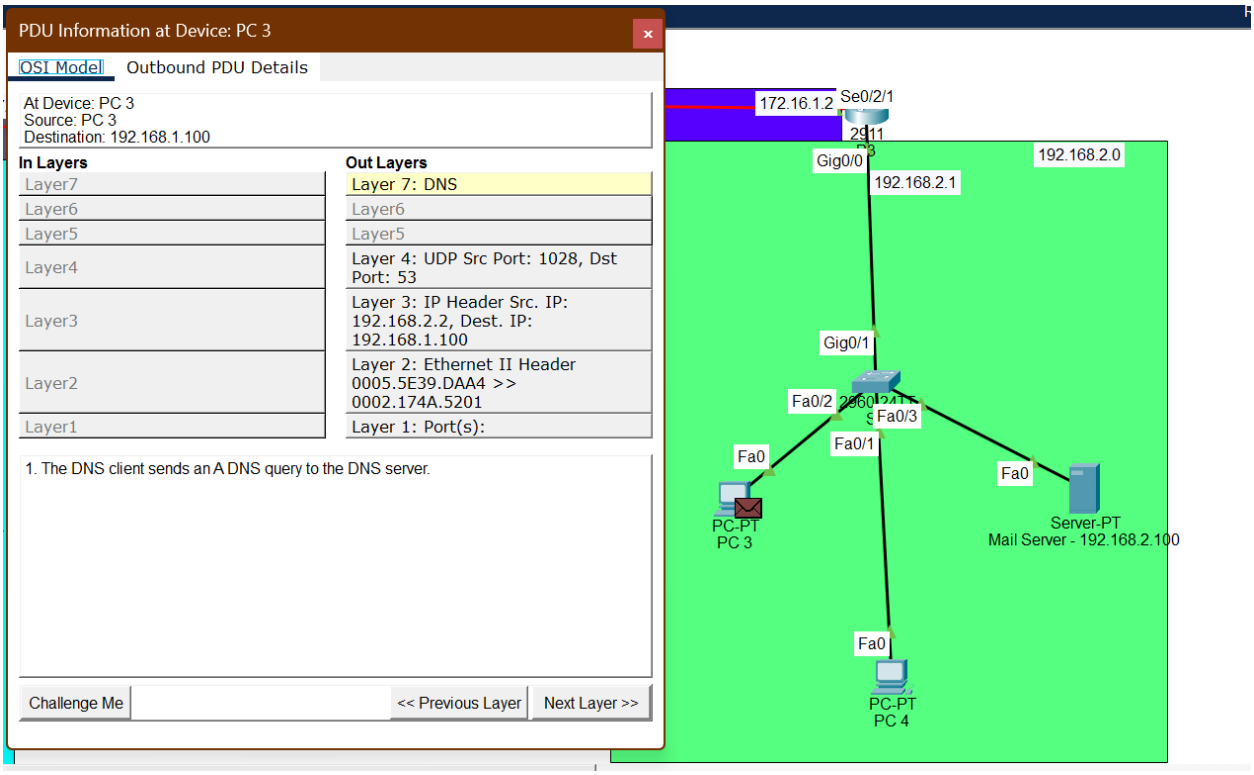
8. Using Simulation Mode, trace an FTP packet from PC-C to the FTP Server and record each hop along the path, noting the Layer 4 details at each step.

First of all, we send the login request to the FTP server. It will search for DNS, and send request to DNS server (192.168.1.100) to find the IP of “ahsan-53143”. Then Server will respond back. Then TCP send the connection request to the FTP server. And if server is idle, then it will respond. After this shake hand original data will be sent. Then the pc will send the login request to server, and then the FTP server will respond back and ask for credentials.

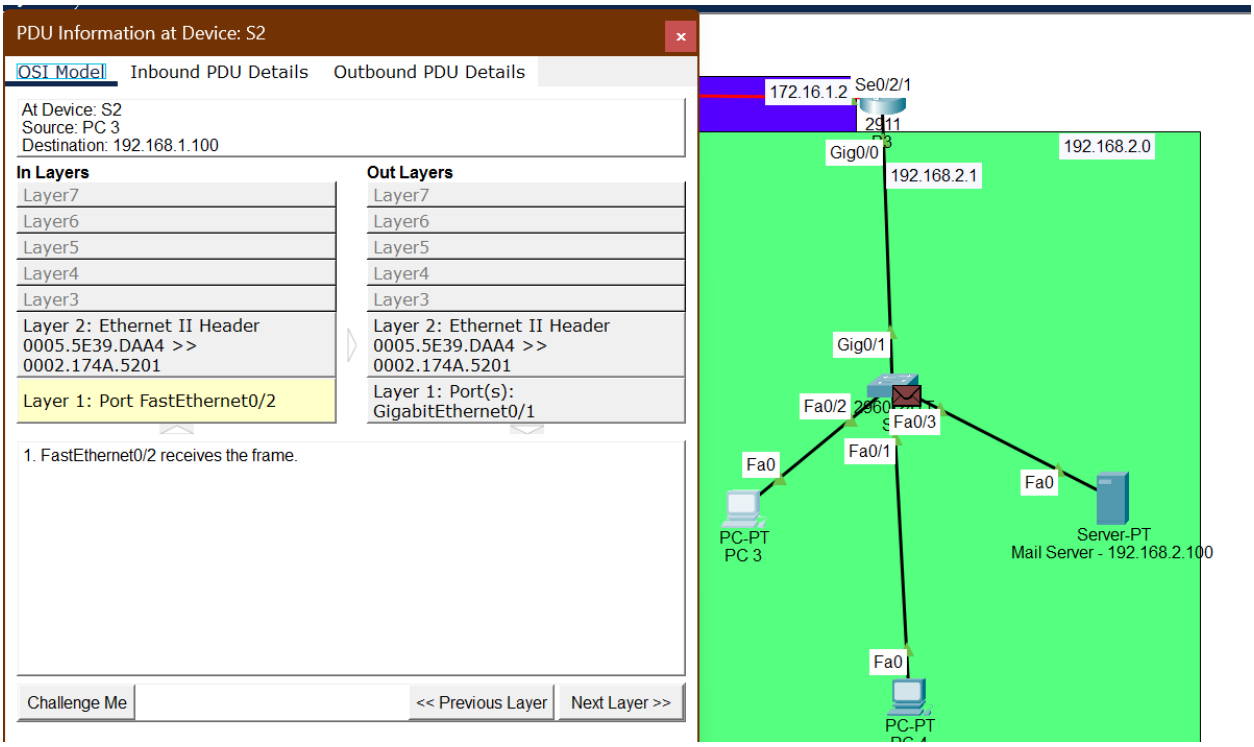
From PC3 send FTP request with a Domain “ahsan-53143”



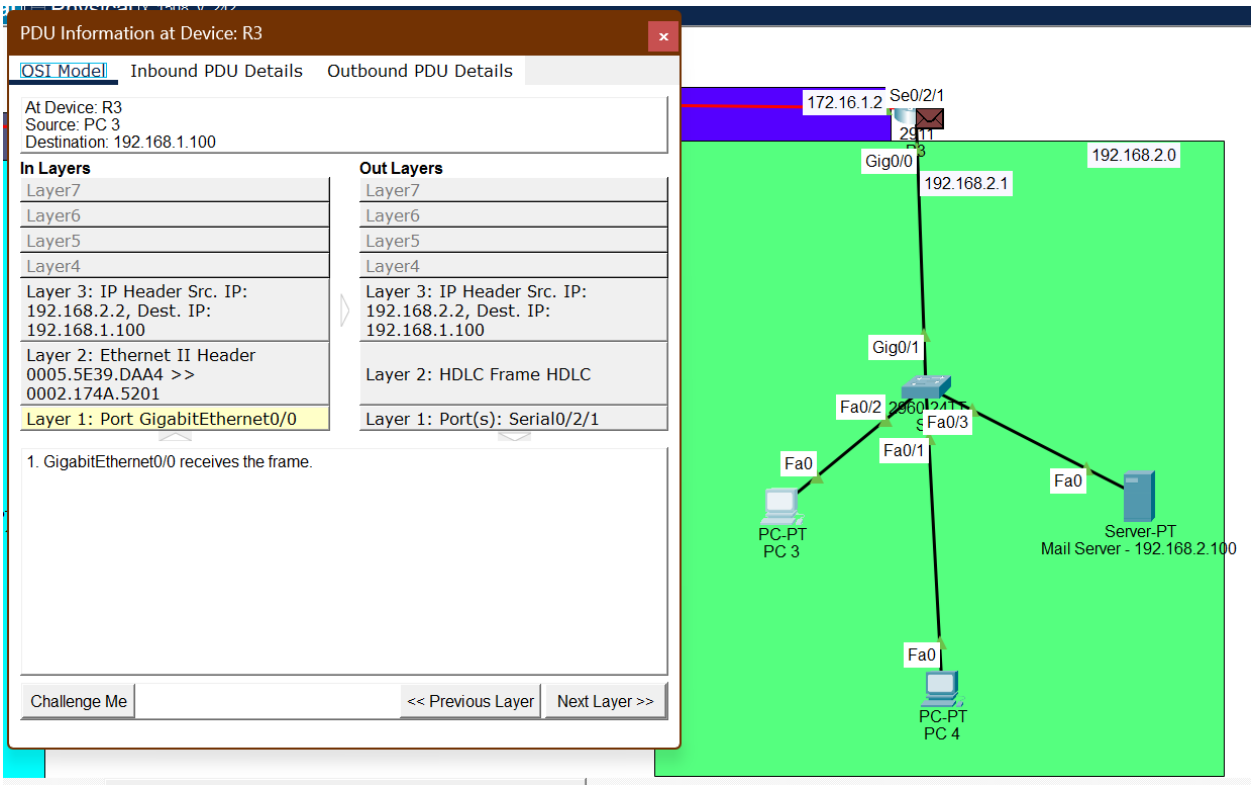
Path from PC 3 to DNS server (192.168.1.100)



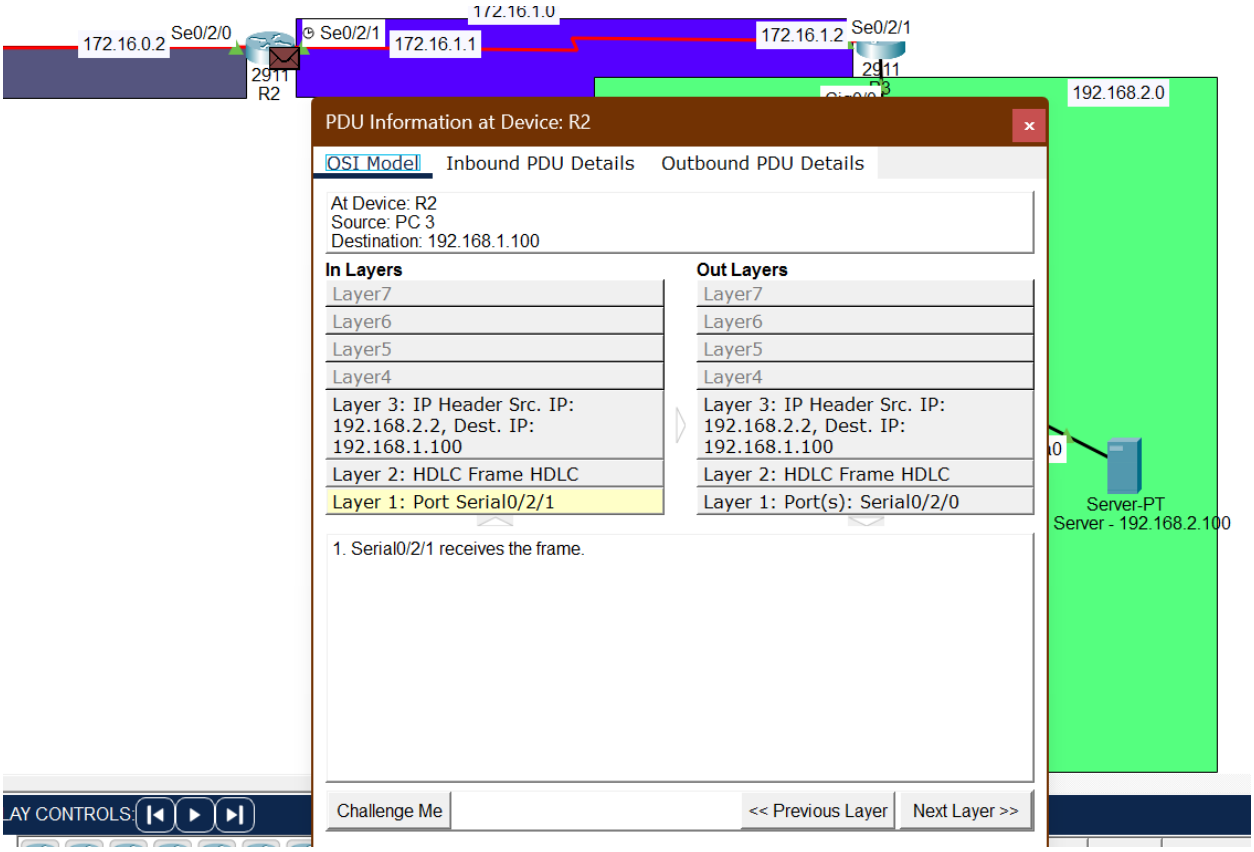
Hop 1: Switch (S2)



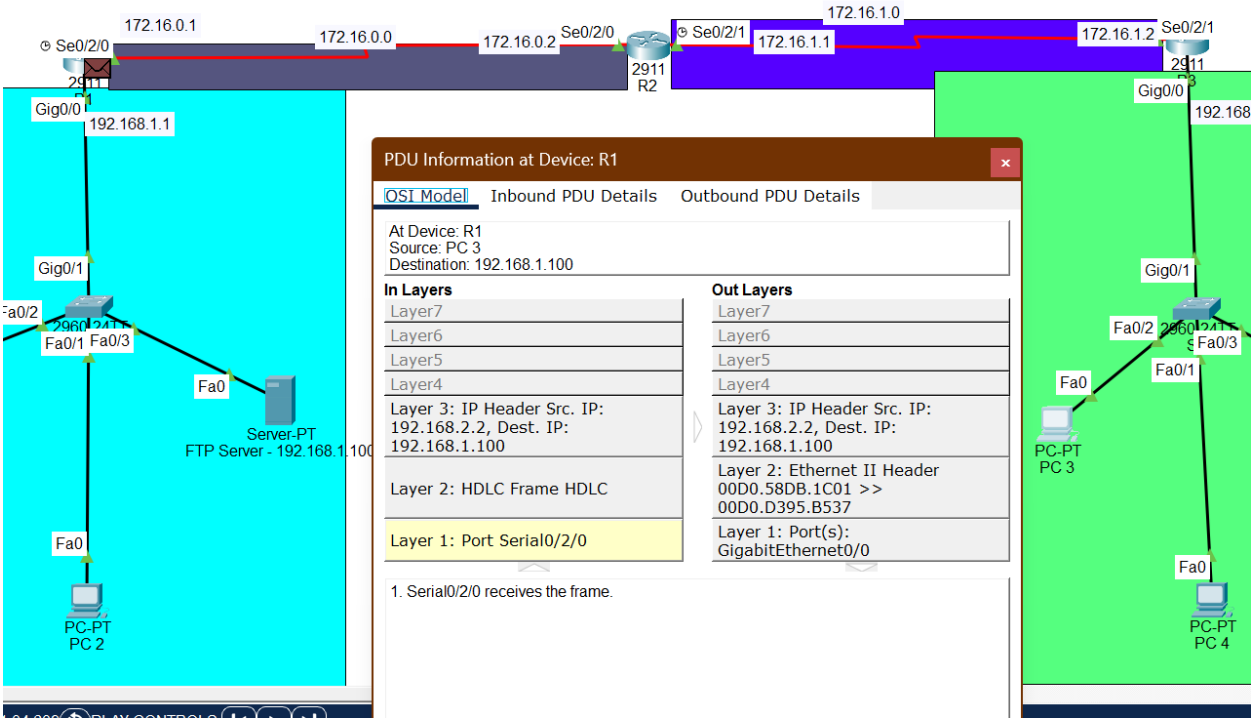
Hop 2: Router (R3)



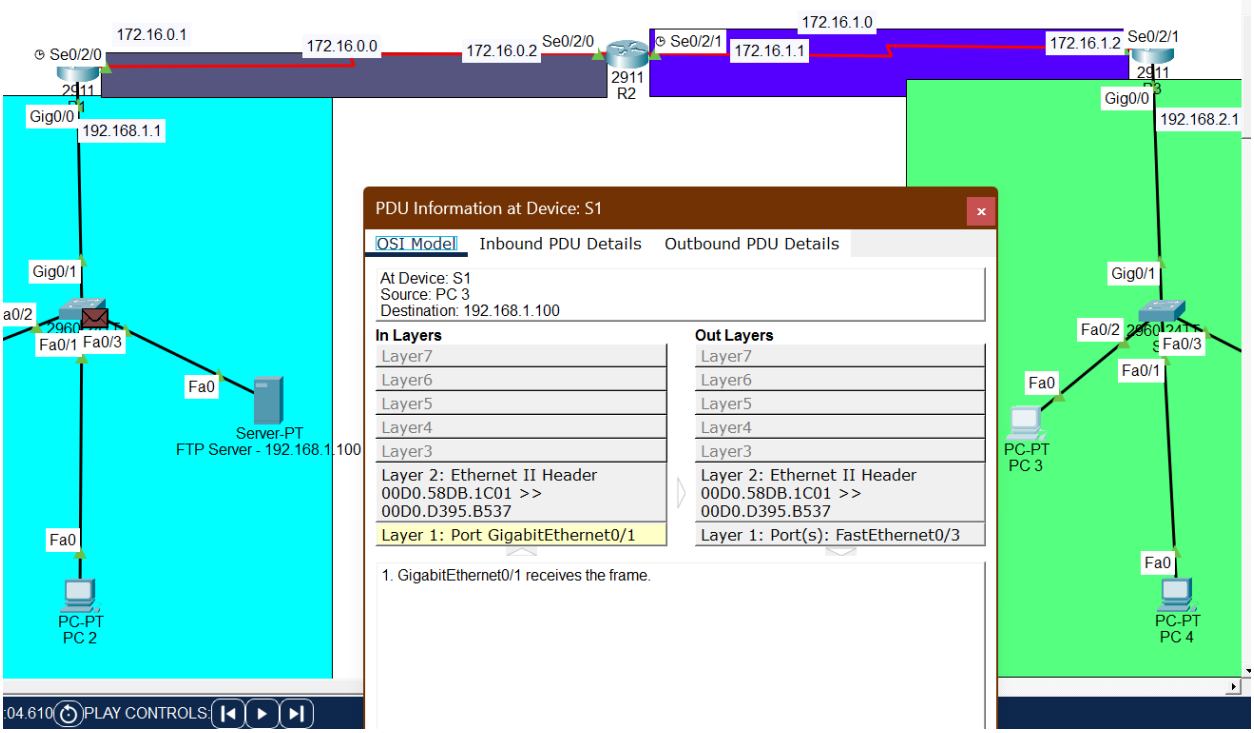
Hop 3: Router (R2)



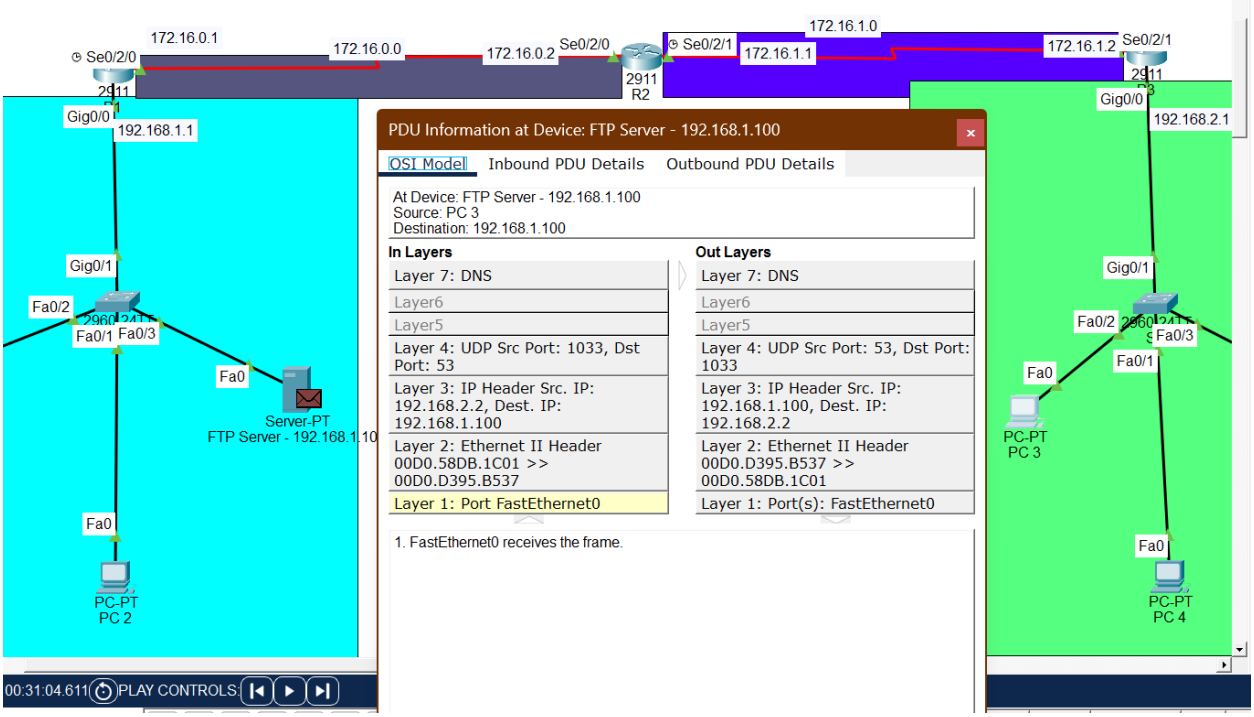
Hop 4: Router (R1)



Hop 5: Router (S1)

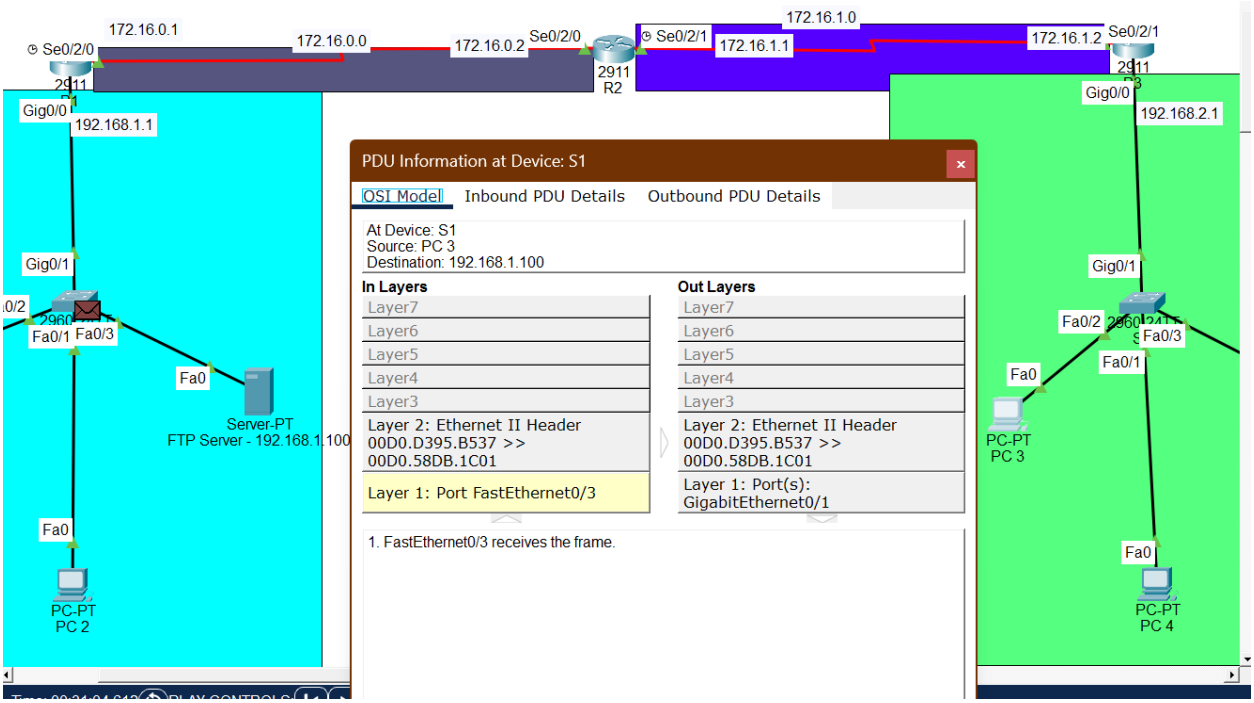


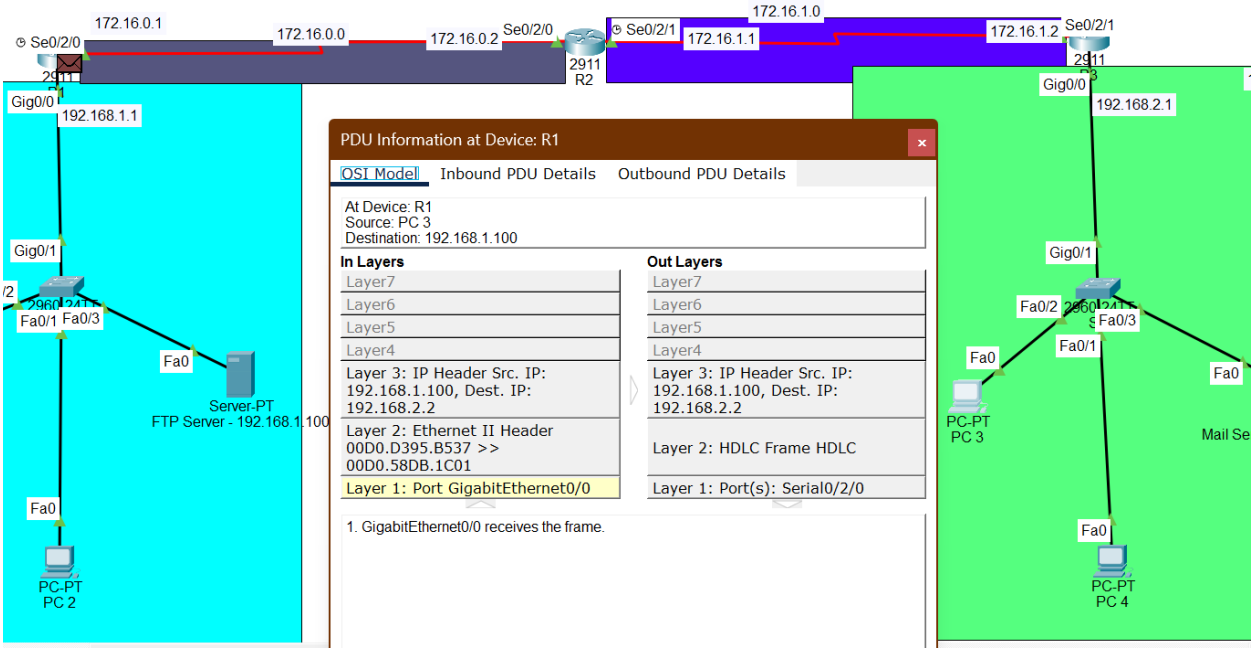
Destination: FTP server

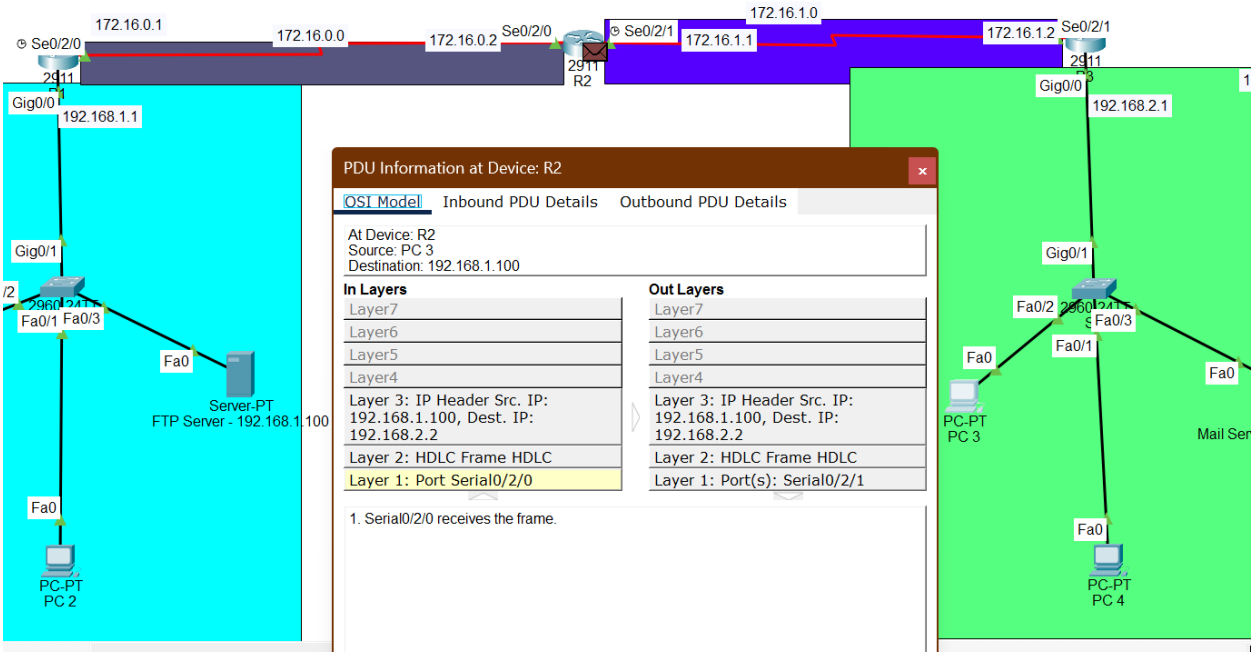


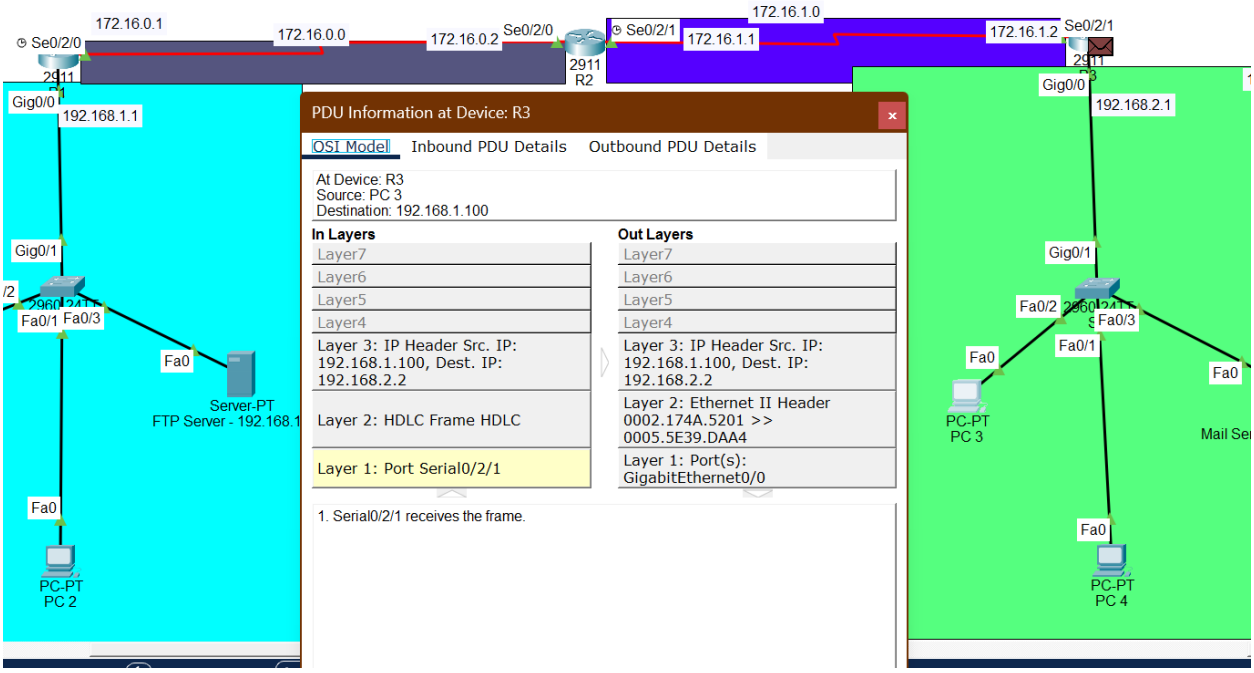
DNS Server Response back to PC 3

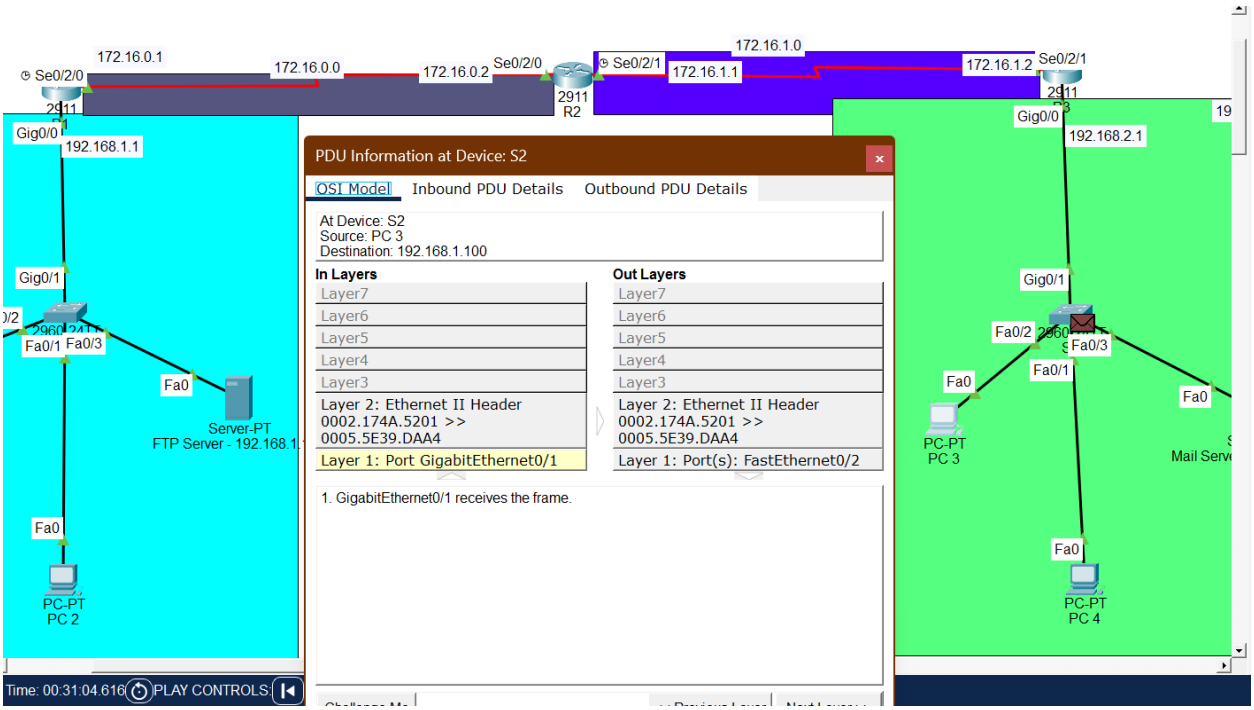
Path of DNS server to PC3

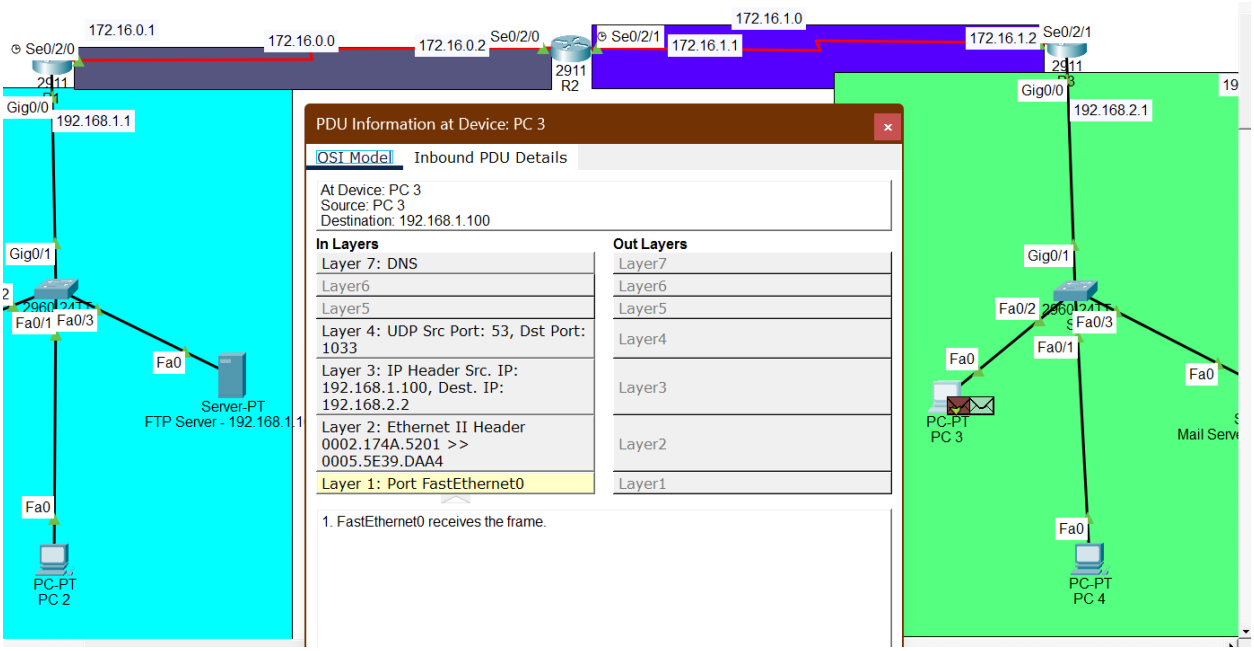

172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
2911
R2
Gig0/0
192.168.1.1
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1.100
PC-PT
PC 2
Gig0/0
192.168.2.1
Gig0/1
Fa0/2
Fa0/3
Fa0/1
PC-PT
PC 3
PC-PT
PC 4
PDU Information at Device: S1
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: S1
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Ethernet II Header
00D0.D395.B537 >>
00D0.58DB.1C01
Layer 1: Port FastEthernet0/3
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Ethernet II Header
00D0.D395.B537 >>
00D0.58DB.1C01
Layer 1: Port(s):
GigabitEthernet0/1
1. FastEthernet0/3 receives the frame.


172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
2911
R2
Gig0/0
192.168.1.1
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1.100
PC-PT
PC 2
Gig0/0
192.168.2.1
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
Mail Se
PC-PT
PC 4
PDU Information at Device: R1
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: R1
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP:
192.168.1.100, Dest. IP:
192.168.2.2
Layer 2: Ethernet II Header
00D0.D395.B537 >>
00D0.58DB.1C01
Layer 1: Port GigabitEthernet0/0
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP:
192.168.1.100, Dest. IP:
192.168.2.2
Layer 2: HDLC Frame HDLC
Layer 1: Port(s): Serial0/2/0
1. GigabitEthernet0/0 receives the frame.


172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
2911
R2
Gig0/0
192.168.1.1
Gig0/0
192.168.2.1
PDU Information at Device: R2
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: R2
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.100, Dest. IP: 192.168.2.2
Layer 2: HDLC Frame HDLC
Layer 1: Port Serial0/2/0
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.100, Dest. IP: 192.168.2.2
Layer 2: HDLC Frame HDLC
Layer 1: Port(s): Serial0/2/1
1. Serial0/2/0 receives the frame.
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1.100
Fa0
PC-PT
PC 2
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
Fa0
Mail Ser
Fa0
PC-PT
PC 4

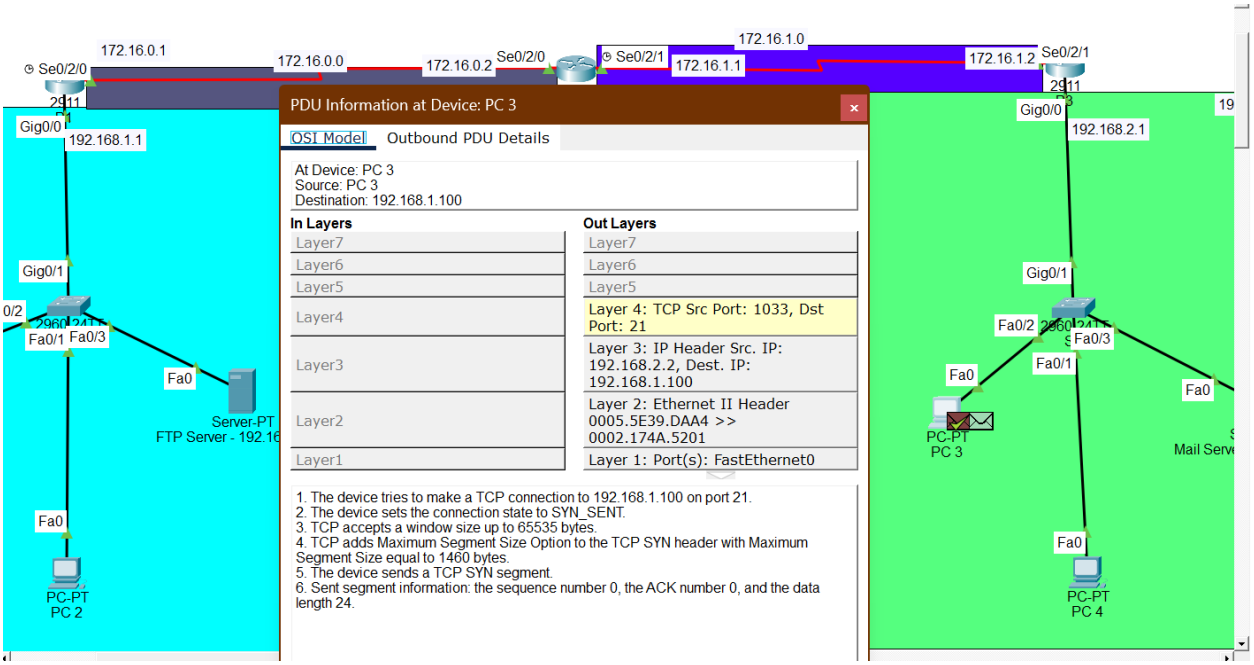

172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
2911
R2
Gig0/0
192.168.1.1
Gig0/0
192.168.2.1
PDU Information at Device: R3
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: R3
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.100, Dest. IP: 192.168.2.2
Layer 2: HDLC Frame HDLC
Layer 1: Port Serial0/2/1
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.100, Dest. IP: 192.168.2.2
Layer 2: Ethernet II Header 0002.174A.5201 >> 0005.5E39.DAA4
Layer 1: Port(s): GigabitEthernet0/0
1. Serial0/2/1 receives the frame.
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1
Fa0
PC-PT
PC 2
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
Fa0
Mail Se
Fa0
PC-PT
PC 4

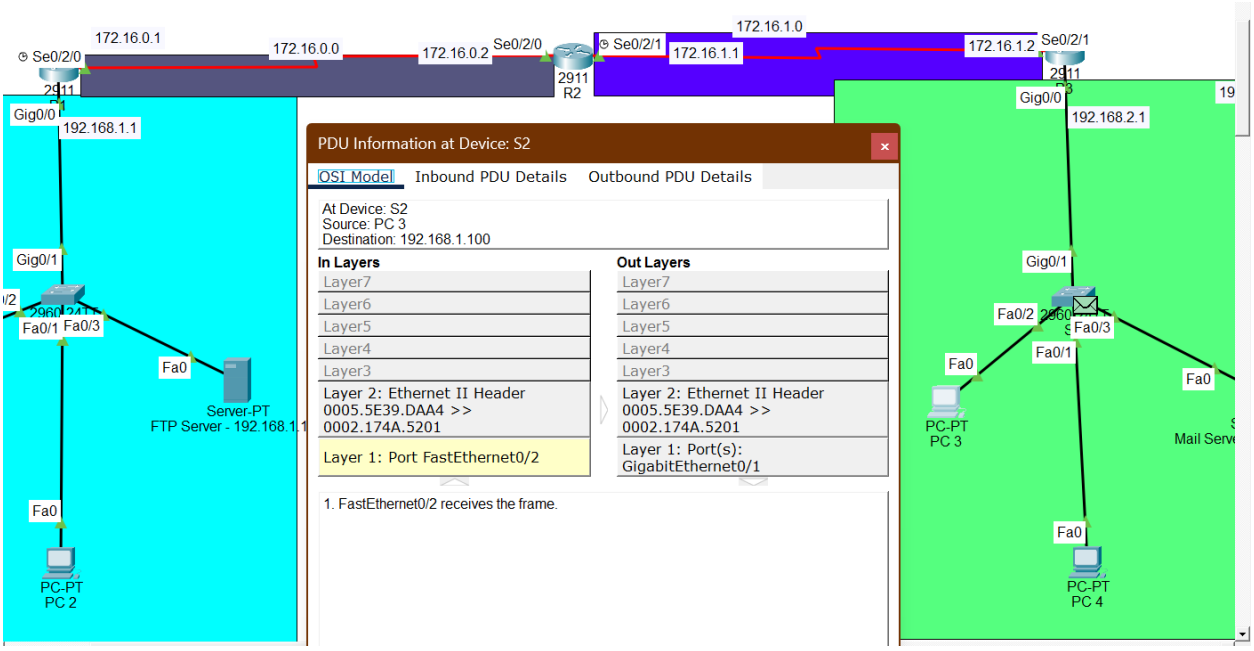


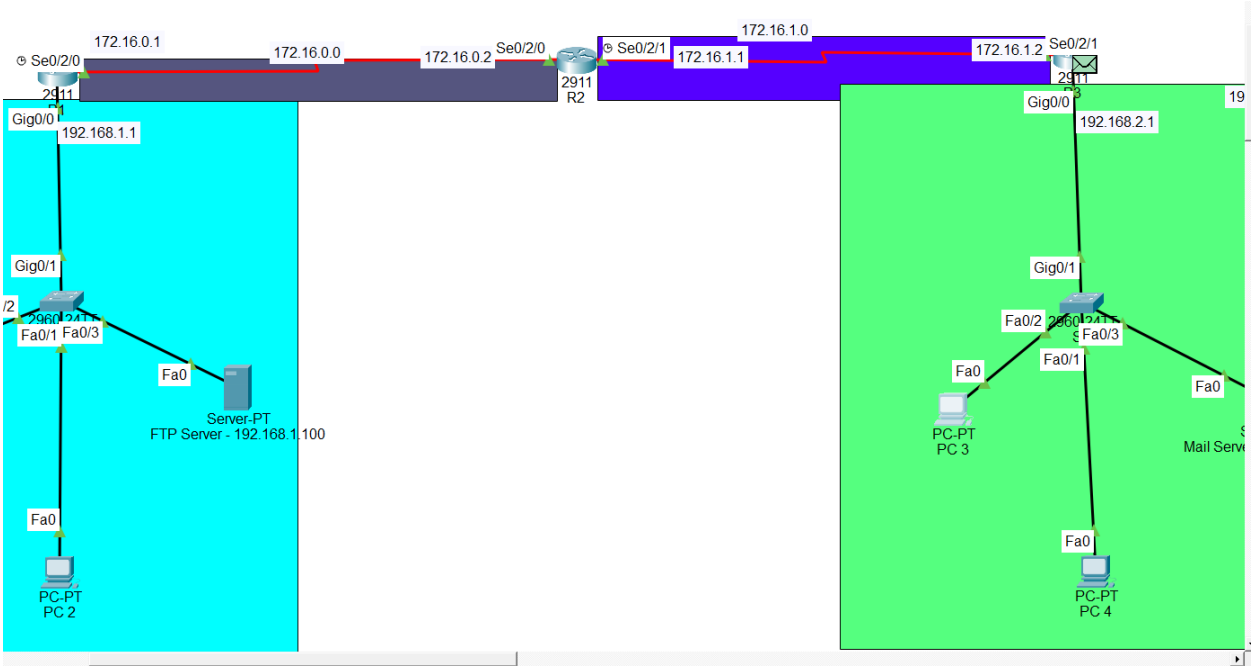


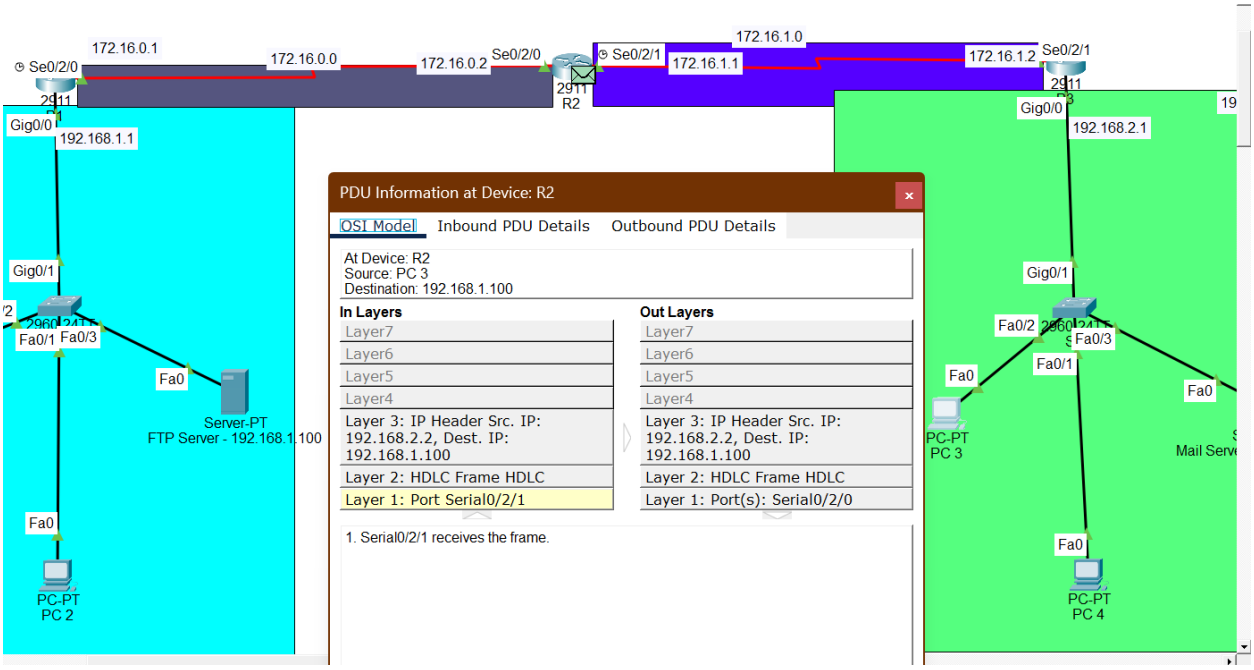
Now It FTP process will start:

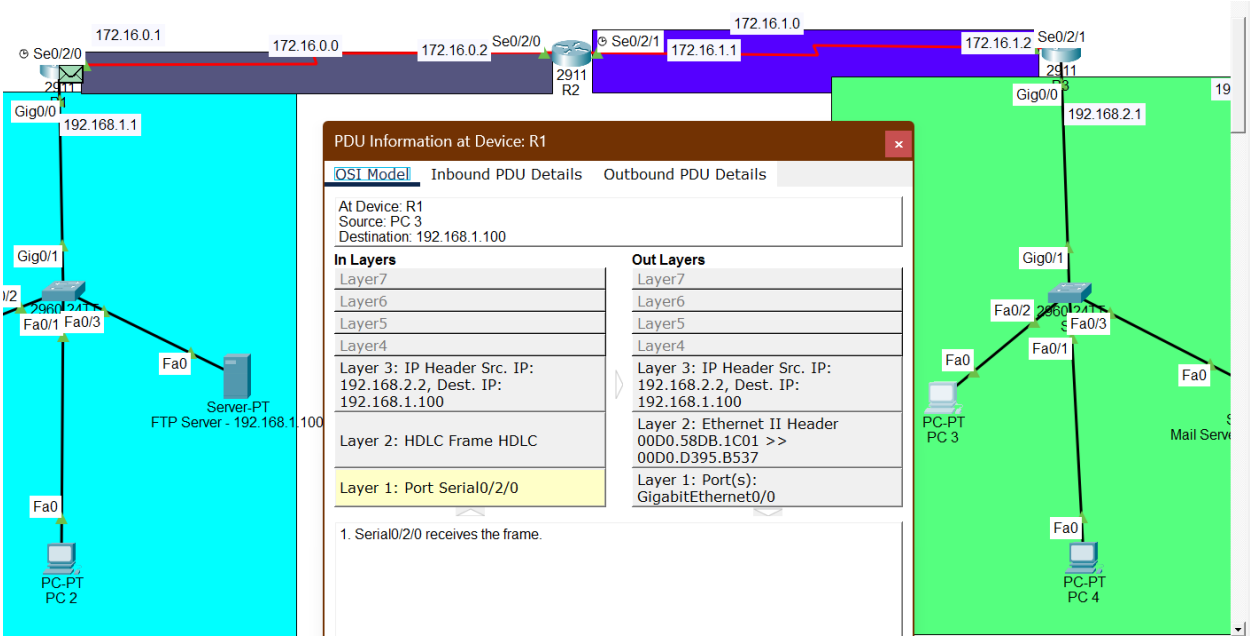
Now going through same hops, it will first shake hand with server and then send the original request to login.

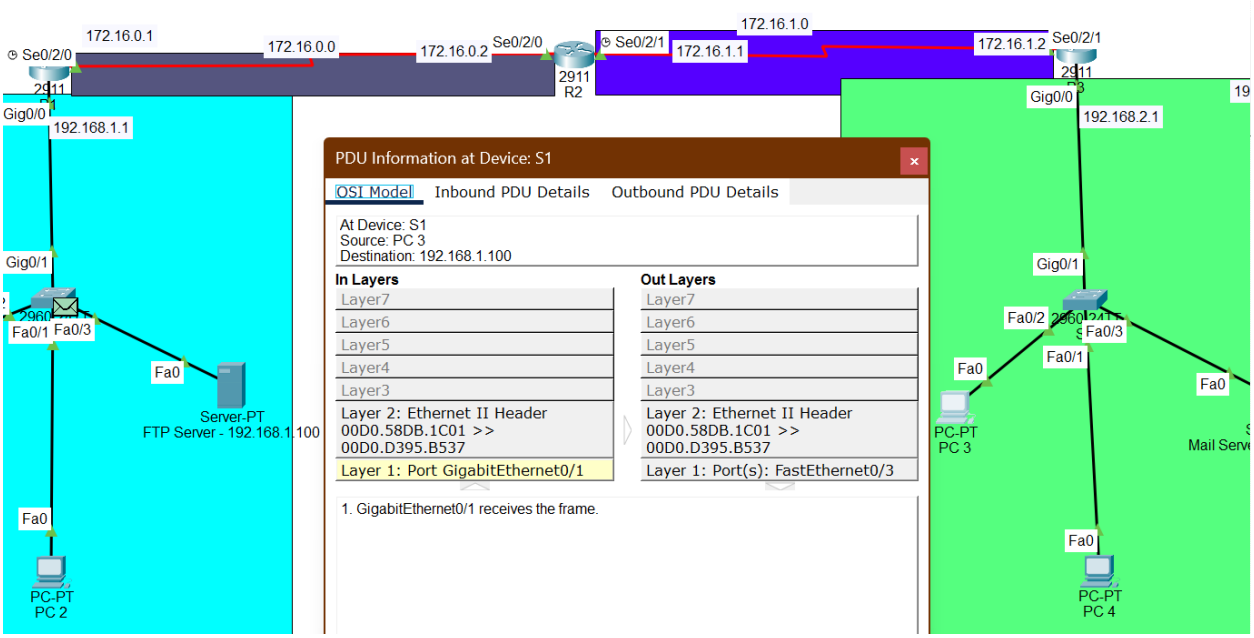

172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
Gig0/0
192.168.1.1
192.168.2.1
Gig0/1
Fa0/1
Fa0/3
Fa0/2
Fa0
Server-PT
FTP Server - 192.16
PC-PT
PC 2
PC-PT
PC 3
PC-PT
PC 4
Mail Serve
PDU Information at Device: PC 3
OSI Model
Outbound PDU Details
At Device: PC 3
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer2
Layer1
Out Layers
Layer7
Layer6
Layer5
Layer 4: TCP Src Port: 1033, Dst Port: 21
Layer 3: IP Header Src. IP: 192.168.2.2, Dest. IP: 192.168.1.100
Layer 2: Ethernet II Header 0005.5E39.DAA4 >> 0002.174A.5201
Layer 1: Port(s): FastEthernet0
1. The device tries to make a TCP connection to 192.168.1.100 on port 21.
2. The device sets the connection state to SYN_SENT.
3. TCP accepts a window size up to 65535 bytes.
4. TCP adds Maximum Segment Size Option to the TCP SYN header with Maximum Segment Size equal to 1460 bytes.
5. The device sends a TCP SYN segment.
6. Sent segment information: the sequence number 0, the ACK number 0, and the data length 24.

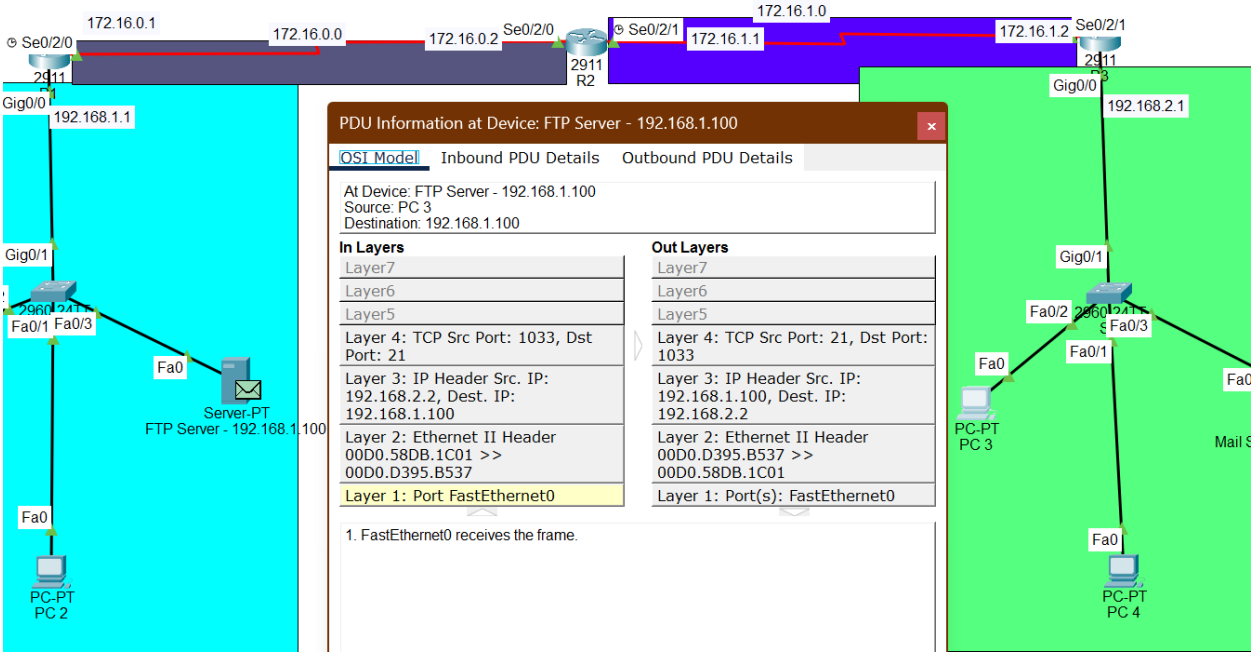

172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
2911
R2
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
Gig0/0
192.168.1.1
192.168.2.1
Gig0/1
Fa0/1
Fa0/3
Fa0/2
Fa0
Server-PT
FTP Server - 192.168.1.1
PC-PT
PC 2
PC-PT
PC 3
PC-PT
PC 4
Mail Serve
PDU Information at Device: S2
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: S2
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Ethernet II Header 0005.5E39.DAA4 >> 0002.174A.5201
Layer 1: Port FastEthernet0/2
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Ethernet II Header 0005.5E39.DAA4 >> 0002.174A.5201
Layer 1: Port(s): GigabitEthernet0/1
1. FastEthernet0/2 receives the frame.


172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
2911
R2
Gig0/0
192.168.1.1
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1.100
PC-PT
PC 2
Gig0/0
192.168.2.1
Gig0/1
Fa0/2
Fa0/3
Fa0/1
PC-PT
PC 3
PC-PT
PC 4

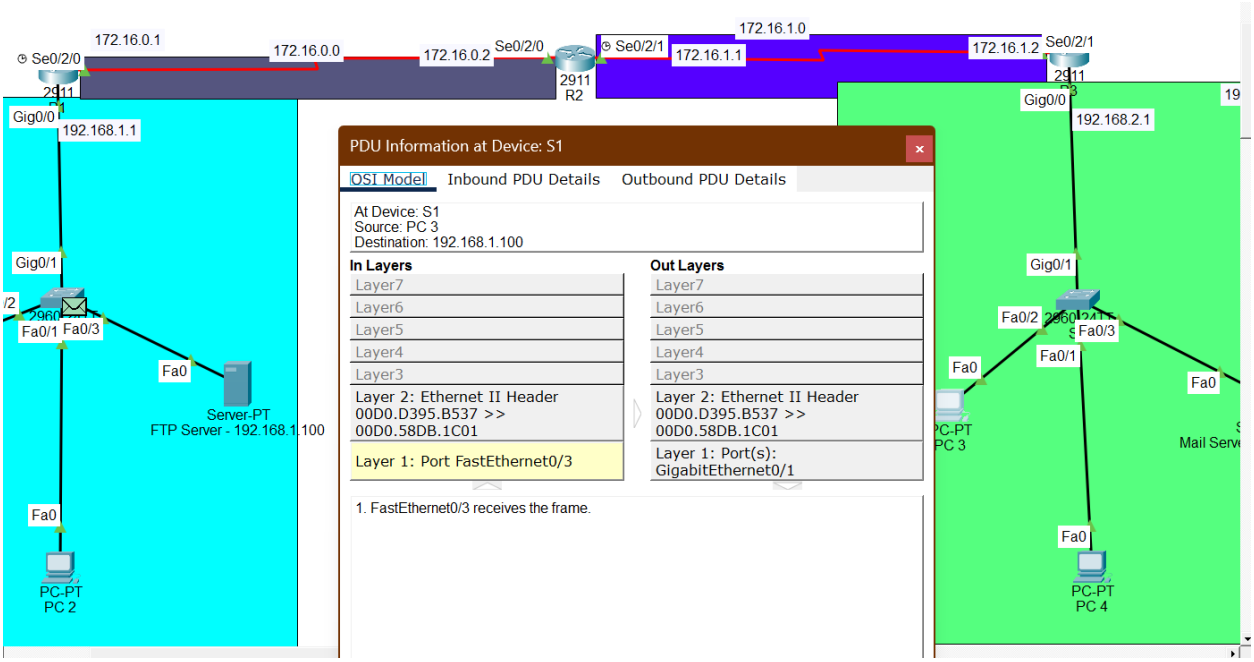

PDU Information at Device: R2
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: R2
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.2.2, Dest. IP: 192.168.1.100
Layer 2: HDLC Frame HDLC
Layer 1: Port Serial0/2/1
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.2.2, Dest. IP: 192.168.1.100
Layer 2: HDLC Frame HDLC
Layer 1: Port(s): Serial0/2/0
1. Serial0/2/1 receives the frame.

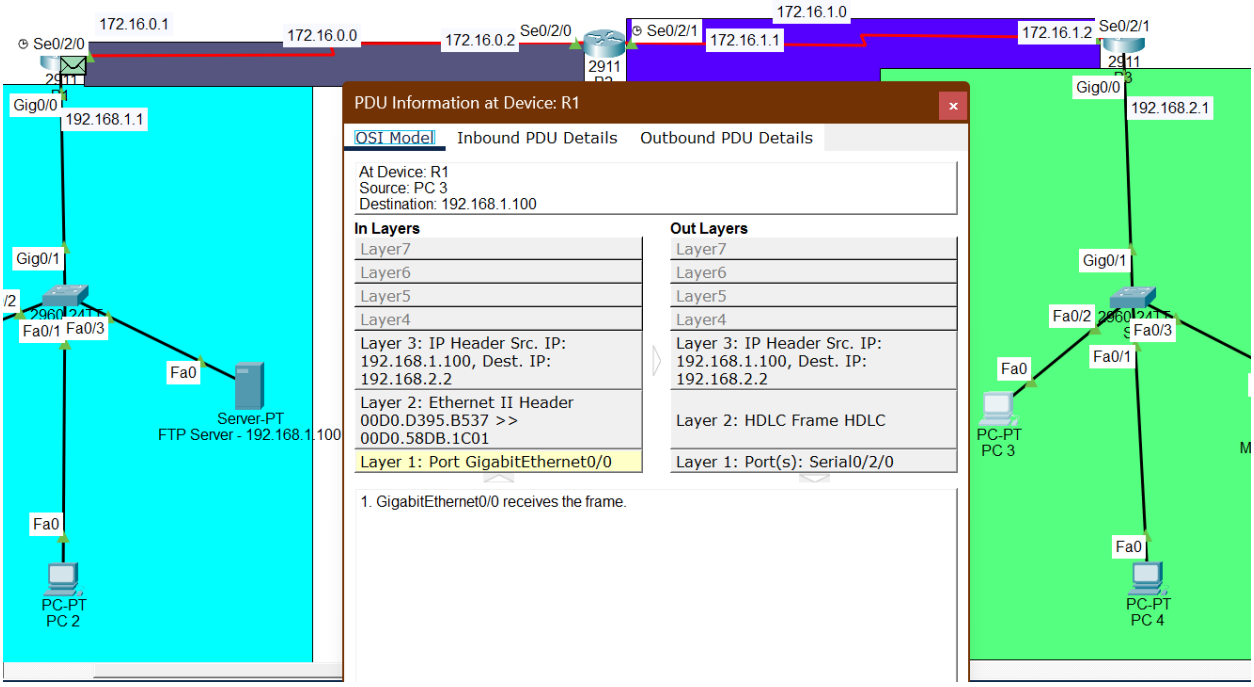

172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
2911
R2
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
Gig0/0
192.168.1.1
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1.100
PC-PT
PC 2
Gig0/0
192.168.2.1
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
Mail Serv
PC-PT
PC 4
PDU Information at Device: R1
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: R1
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.2.2, Dest. IP: 192.168.1.100
Layer 2: HDLC Frame HDLC
Layer 1: Port Serial0/2/0
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.2.2, Dest. IP: 192.168.1.100
Layer 2: Ethernet II Header 00D0.58DB.1C01 >> 00D0.D395.B537
Layer 1: Port(s): GigabitEthernet0/0
1. Serial0/2/0 receives the frame.

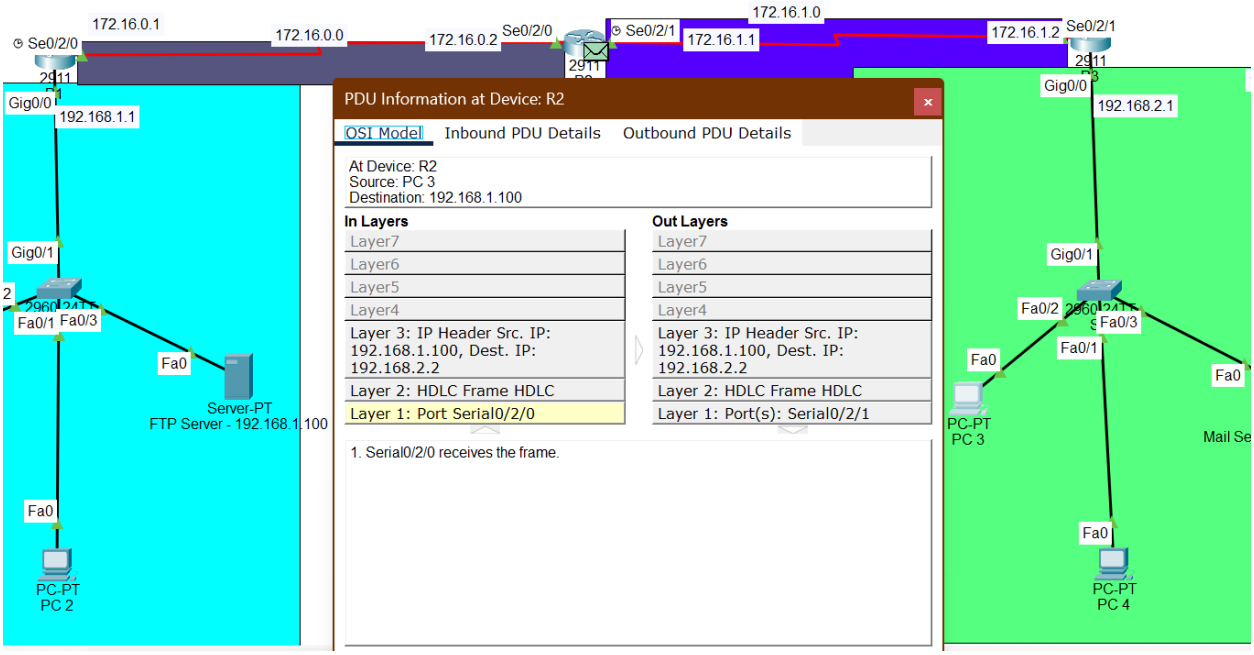

172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
2911
R2
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
Gig0/0
192.168.1.1
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1.100
PC-PT
PC 2
Gig0/0
192.168.2.1
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
Mail Serv
PC-PT
PC 4
PDU Information at Device: S1
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: S1
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Ethernet II Header 00D0.58DB.1C01 >> 00D0.D395.B537
Layer 1: Port GigabitEthernet0/1
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Ethernet II Header 00D0.58DB.1C01 >> 00D0.D395.B537
Layer 1: Port(s): FastEthernet0/3
1. GigabitEthernet0/1 receives the frame.

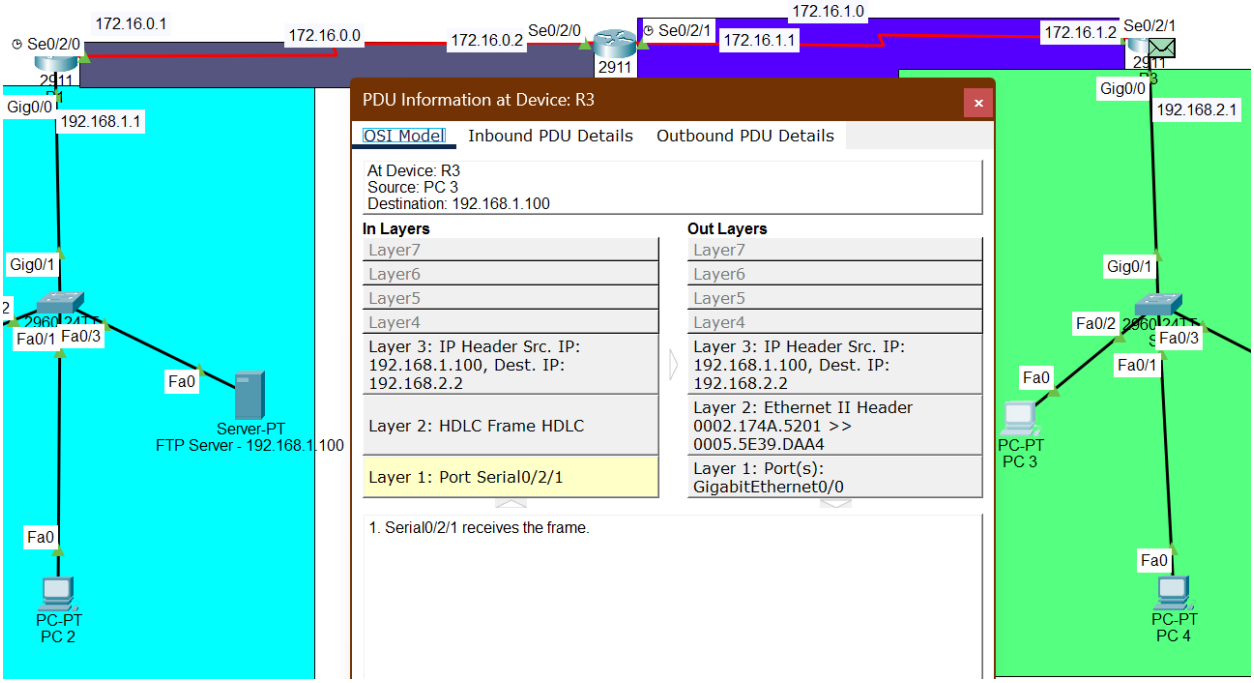


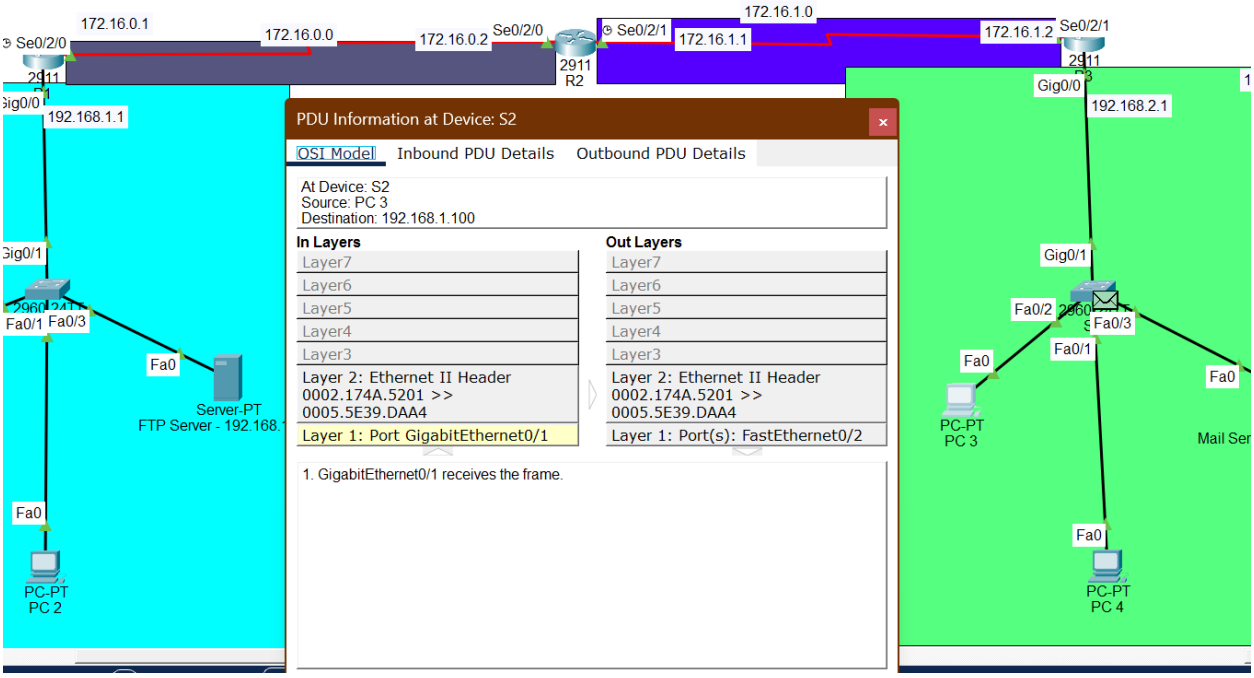
Now server will response back that I am ready you can send the data.

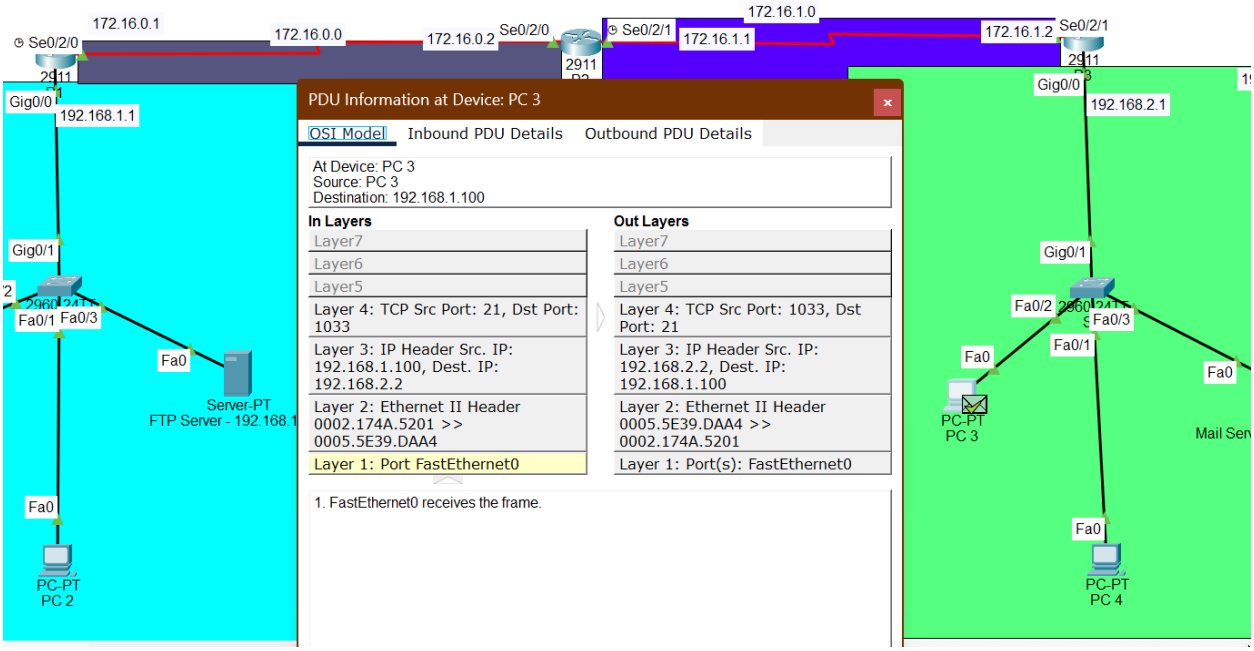



172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
Gig0/0
192.168.1.1
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1.100
Fa0
PC-PT
PC 2
PDU Information at Device: R1
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: R1
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.100, Dest. IP: 192.168.2.2
Layer 2: Ethernet II Header 00D0.D395.B537 >> 00D0.58DB.1C01
Layer 1: Port GigabitEthernet0/0
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.100, Dest. IP: 192.168.2.2
Layer 2: HDLC Frame HDLC
Layer 1: Port(s): Serial0/2/0
1. GigabitEthernet0/0 receives the frame.
Gig0/0
192.168.2.1
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
Fa0
PC-PT
PC 4

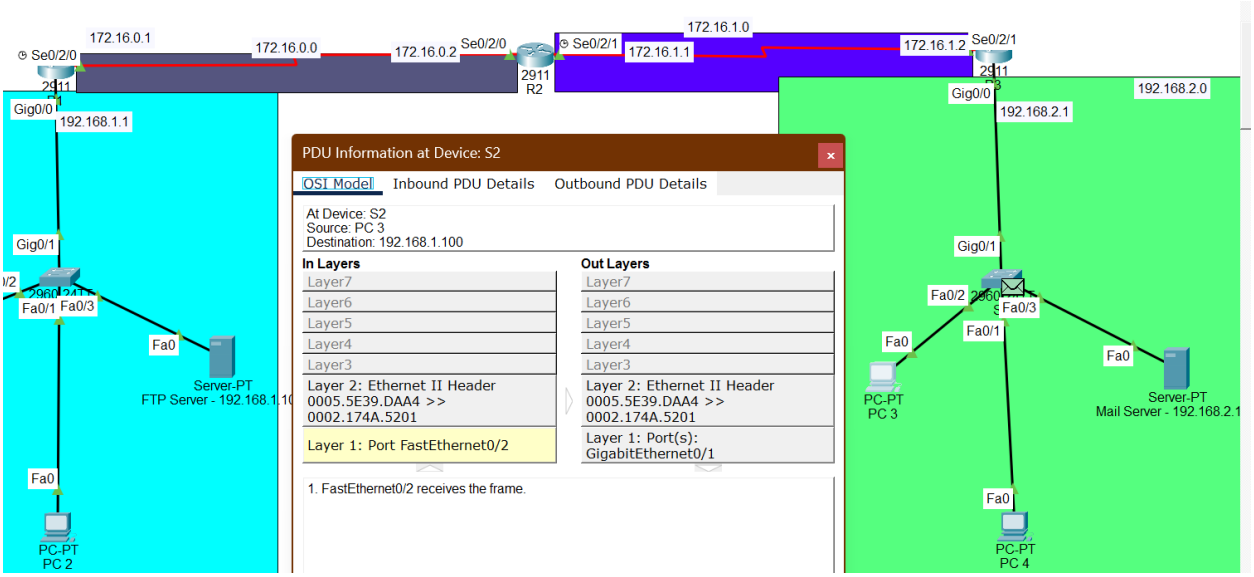

172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
Gig0/0
192.168.1.1
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1.100
Fa0
PC-PT
PC 2
PDU Information at Device: R2
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: R2
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.100, Dest. IP: 192.168.2.2
Layer 2: HDLC Frame HDLC
Layer 1: Port Serial0/2/0
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.100, Dest. IP: 192.168.2.2
Layer 2: HDLC Frame HDLC
Layer 1: Port(s): Serial0/2/1
1. Serial0/2/0 receives the frame.
Gig0/0
192.168.2.1
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
Fa0
Fa0
PC-PT
PC 4

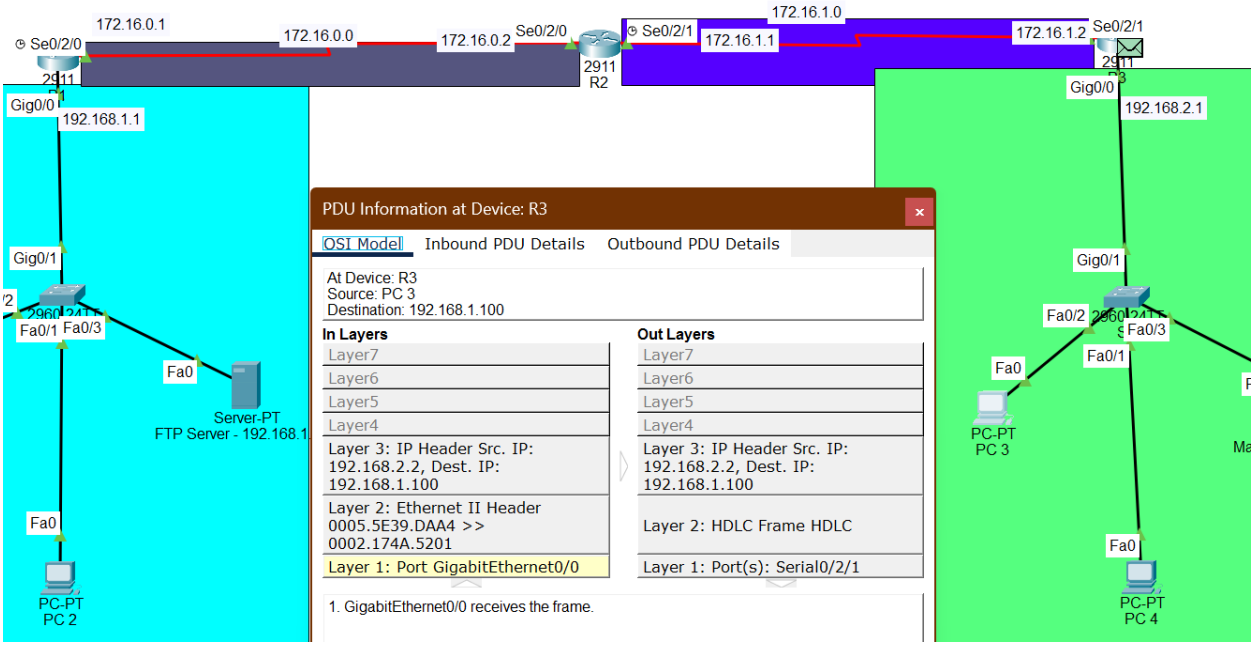

172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
2911
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
Gig0/0
192.168.1.1
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1.100
Fa0
PC-PT
PC 2
PDU Information at Device: R3
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: R3
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.100, Dest. IP: 192.168.2.2
Layer 2: HDLC Frame HDLC
Layer 1: Port Serial0/2/1
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.100, Dest. IP: 192.168.2.2
Layer 2: Ethernet II Header 0002.174A.5201 >> 0005.5E39.DAA4
Layer 1: Port(s): GigabitEthernet0/0
1. Serial0/2/1 receives the frame.
Gig0/0
192.168.2.1
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
Fa0
PC-PT
PC 4

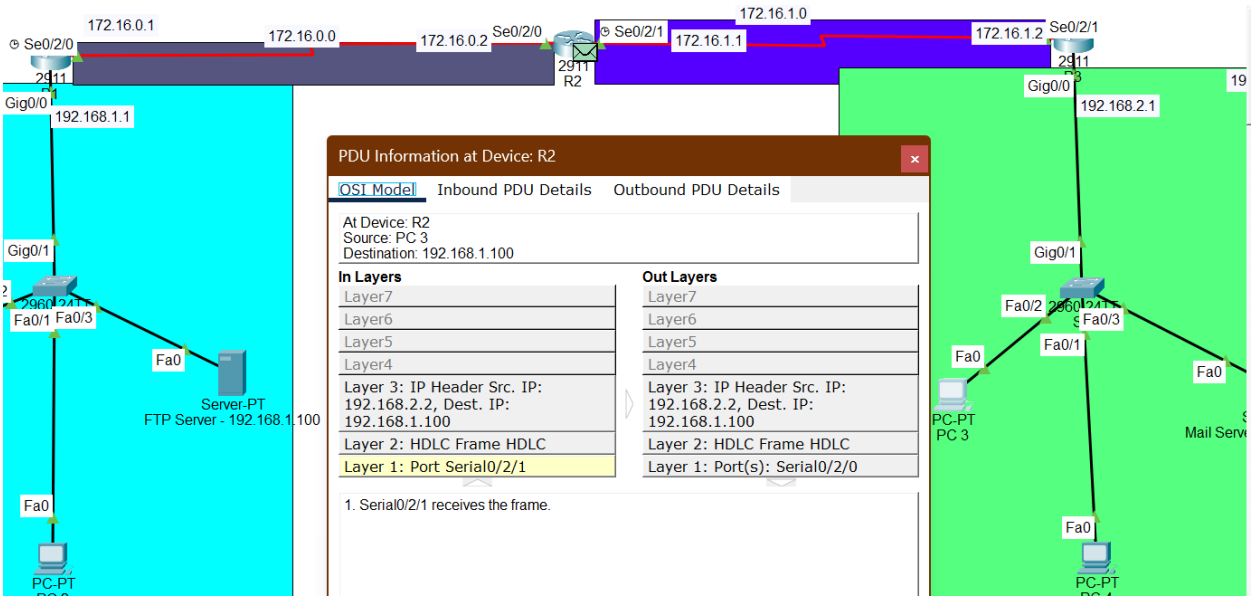

172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
2911
R2
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
192.168.1.1
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1
Fa0
PC-PT
PC 2
PDU Information at Device: S2
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: S2
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Ethernet II Header 0002.174A.5201 >> 0005.5E39.DAA4
Layer 1: Port GigabitEthernet0/1
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Ethernet II Header 0002.174A.5201 >> 0005.5E39.DAA4
Layer 1: Port(s): FastEthernet0/2
1. GigabitEthernet0/1 receives the frame.
Gig0/0
192.168.2.1
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
Fa0
Mail Ser
Fa0
PC-PT
PC 4

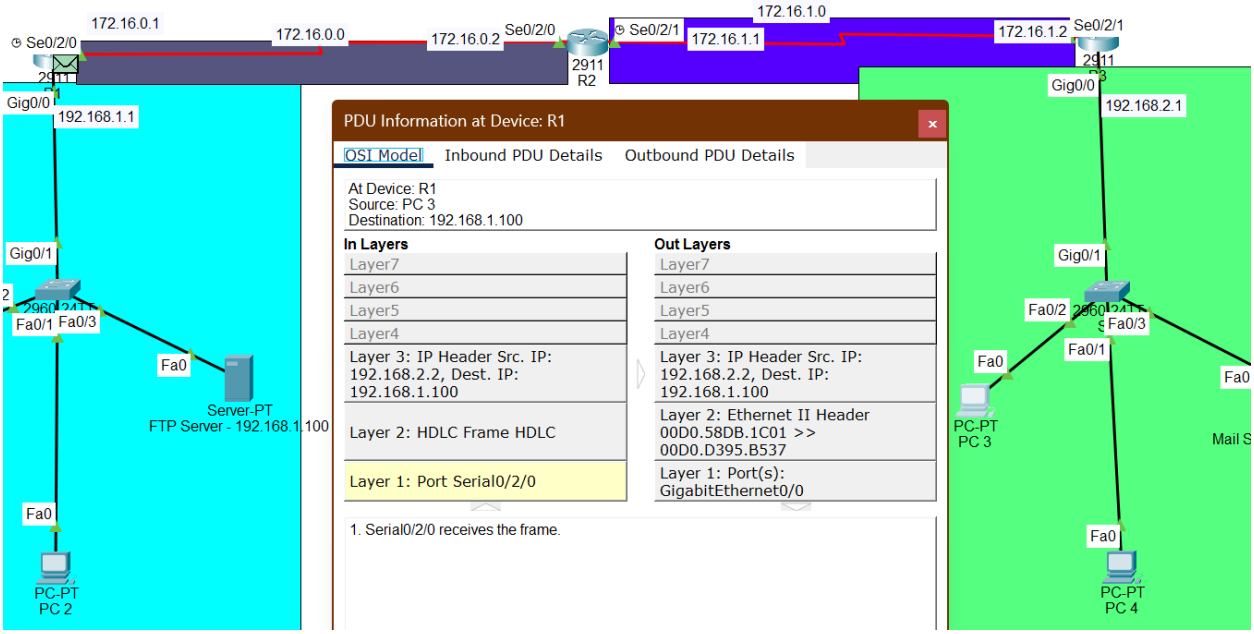


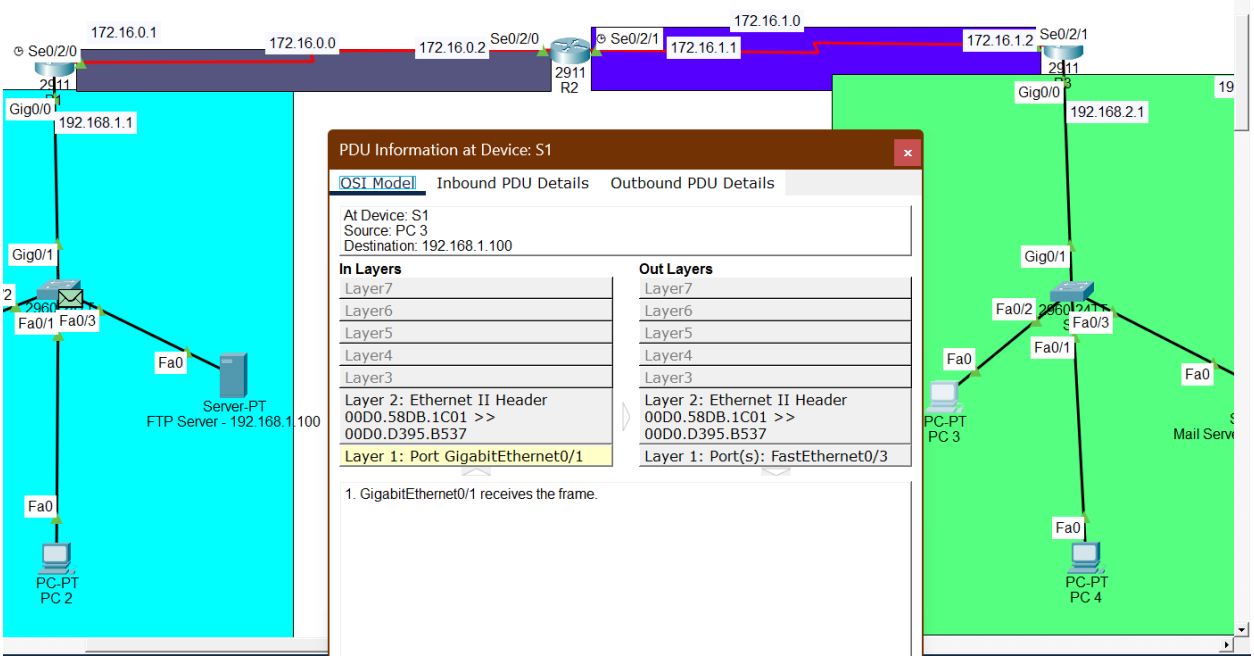
Now the PC will send the login request.

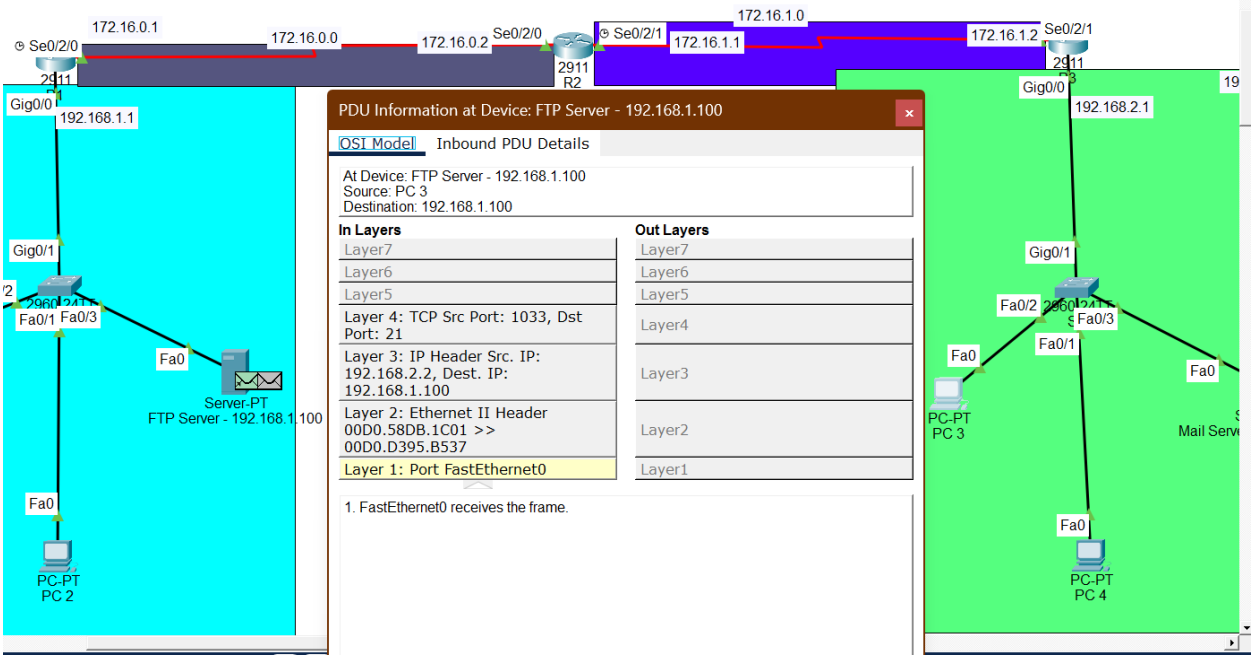



172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
2911
R2
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
Gig0/0
192.168.1.1
Gig0/0
192.168.2.1
PDU Information at Device: R3
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: R3
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.2.2, Dest. IP: 192.168.1.100
Layer 2: Ethernet II Header 0005.5E39.DAA4 >> 0002.174A.5201
Layer 1: Port GigabitEthernet0/0
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.2.2, Dest. IP: 192.168.1.100
Layer 2: HDLC Frame HDLC
Layer 1: Port(s): Serial0/2/1
1. GigabitEthernet0/0 receives the frame.
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1
PC-PT
PC 2
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
PC-PT
PC 4

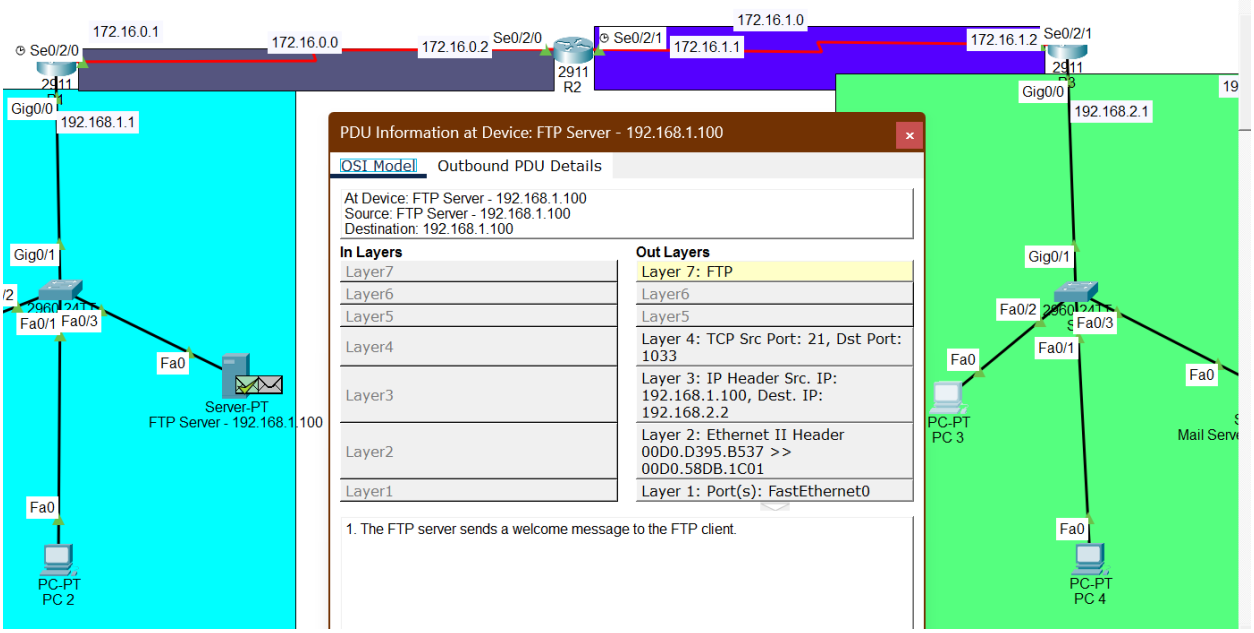

172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
2911
R2
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
Gig0/0
192.168.1.1
Gig0/0
192.168.2.1
PDU Information at Device: R2
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: R2
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.2.2, Dest. IP: 192.168.1.100
Layer 2: HDLC Frame HDLC
Layer 1: Port Serial0/2/1
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.2.2, Dest. IP: 192.168.1.100
Layer 2: HDLC Frame HDLC
Layer 1: Port(s): Serial0/2/0
1. Serial0/2/1 receives the frame.
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1.100
Fa0
PC-PT
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
Fa0
Fa0
PC-PT

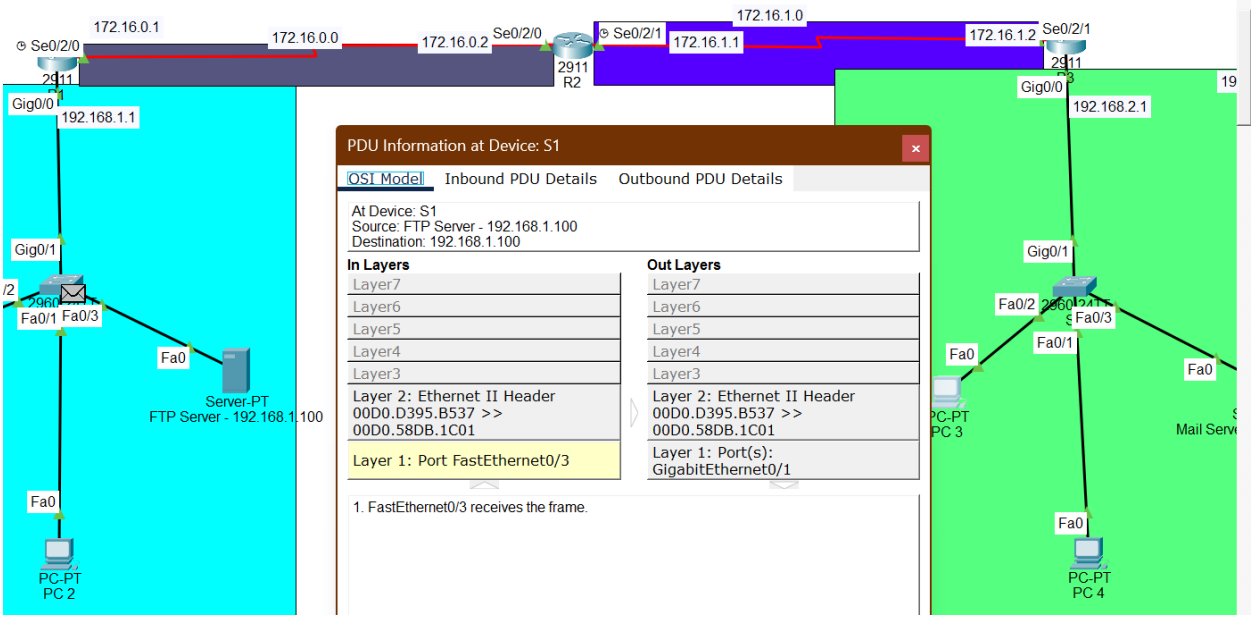

172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
2911
R2
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
Gig0/0
192.168.1.1
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1.100
Fa0
PC-PT
PC 2
Gig0/0
192.168.2.1
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
Fa0
Mail S
Fa0
PC-PT
PC 4
PDU Information at Device: R1
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: R1
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.2.2, Dest. IP: 192.168.1.100
Layer 2: HDLC Frame HDLC
Layer 1: Port Serial0/2/0
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.2.2, Dest. IP: 192.168.1.100
Layer 2: Ethernet II Header 00D0.58DB.1C01 >> 00D0.D395.B537
Layer 1: Port(s): GigabitEthernet0/0
1. Serial0/2/0 receives the frame.

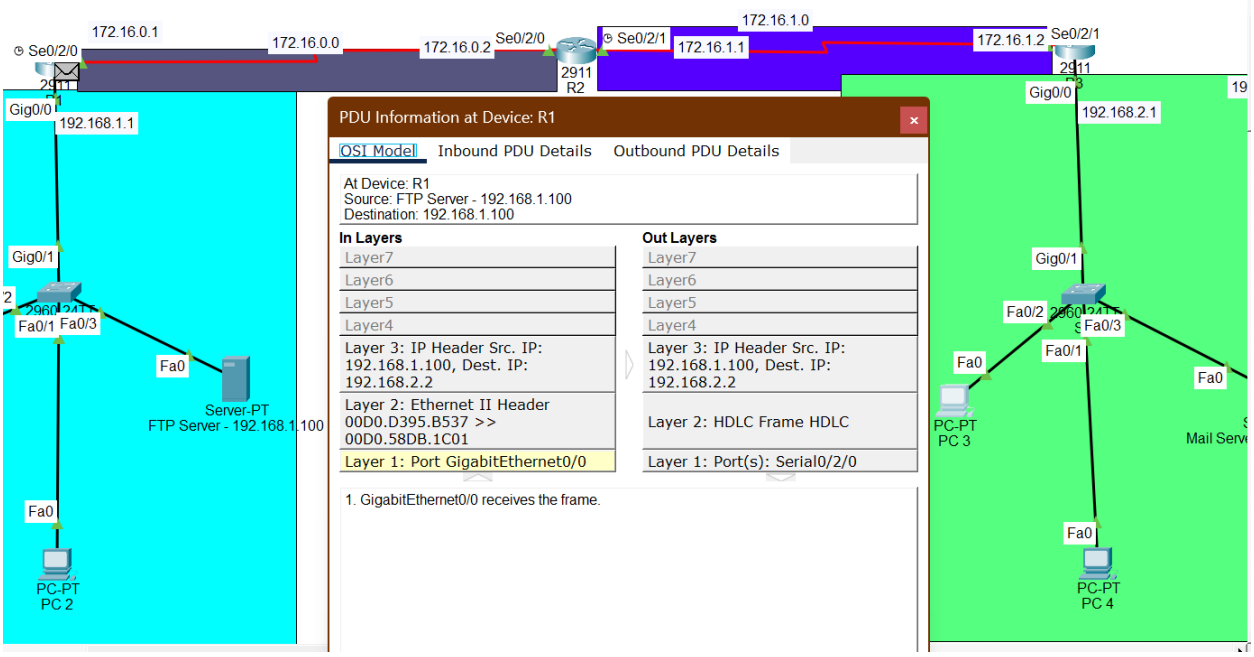

172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
2911
R2
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
Gig0/0
192.168.1.1
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1.100
Fa0
PC-PT
PC 2
Gig0/0
192.168.2.1
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
Fa0
Mail Serve
Fa0
PC-PT
PC 4
PDU Information at Device: S1
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: S1
Source: PC 3
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Ethernet II Header 00D0.58DB.1C01 >> 00D0.D395.B537
Layer 1: Port GigabitEthernet0/1
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Ethernet II Header 00D0.58DB.1C01 >> 00D0.D395.B537
Layer 1: Port(s): FastEthernet0/3
1. GigabitEthernet0/1 receives the frame.

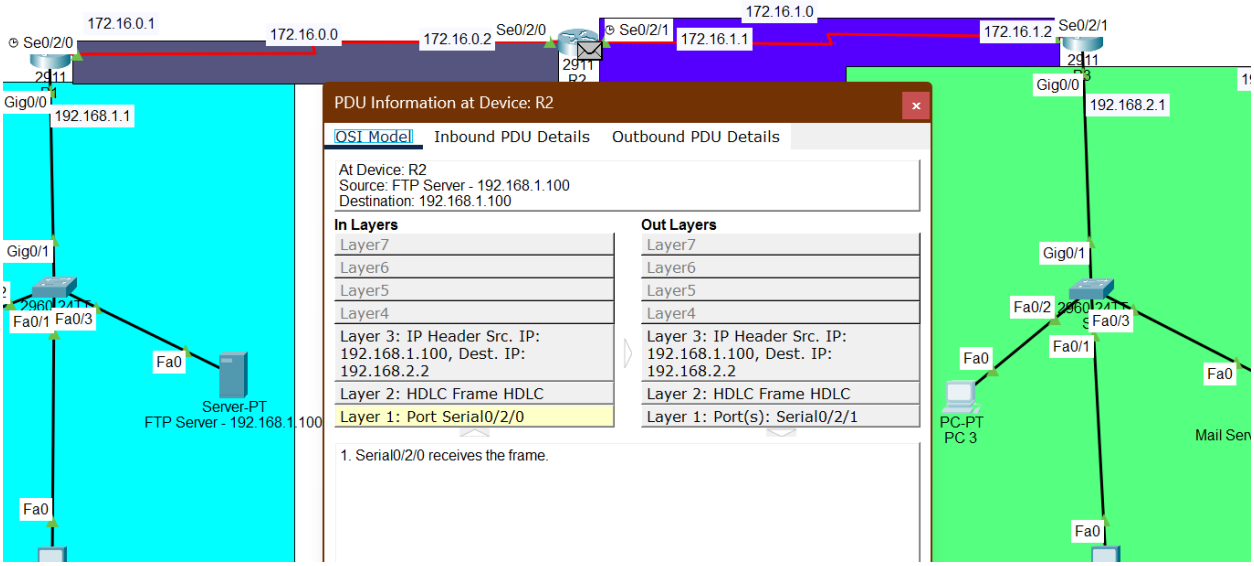


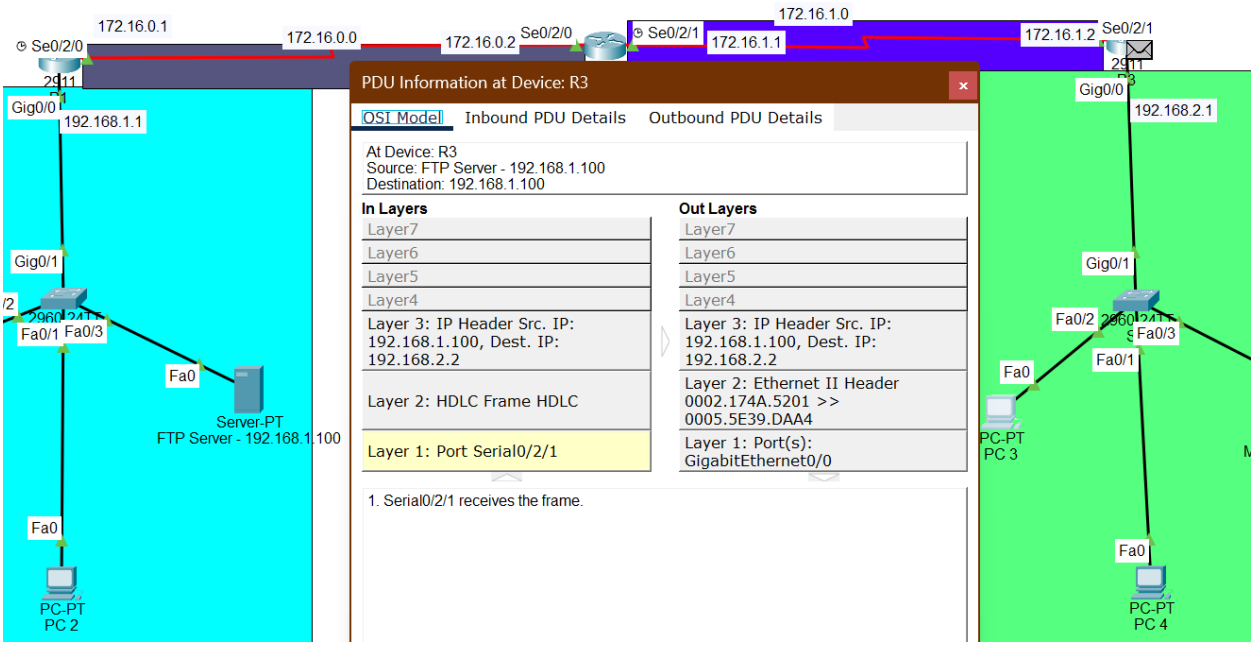
Now the server has received login and then it will ask for username back form the client.

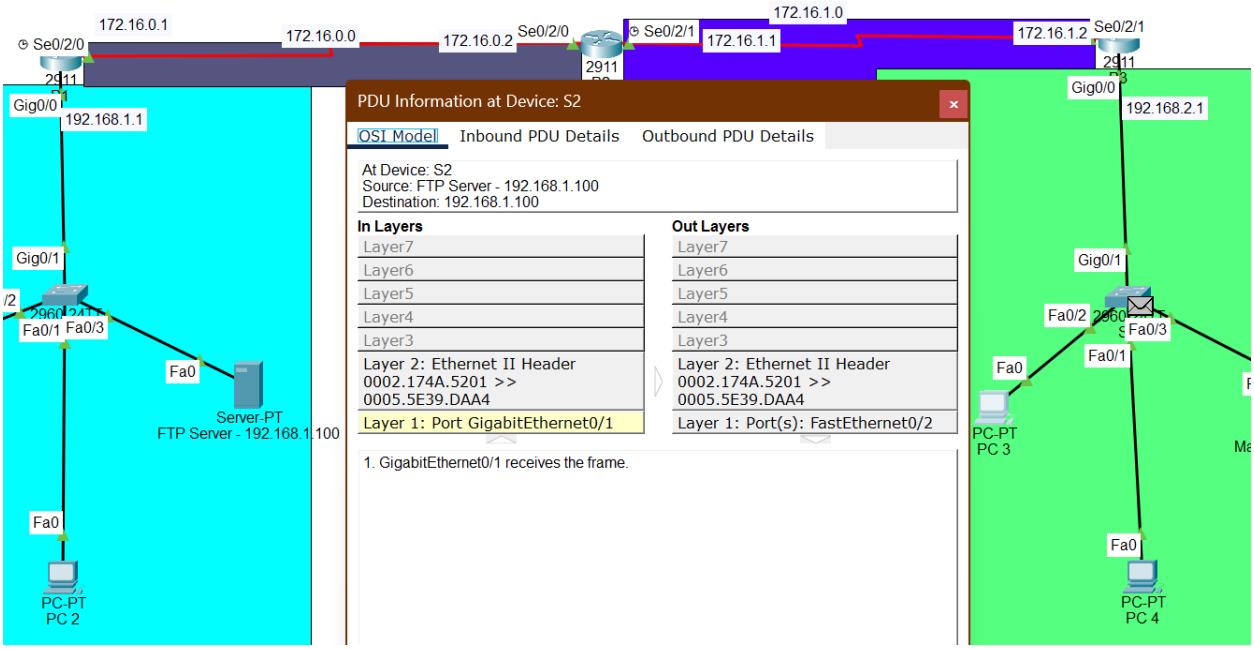


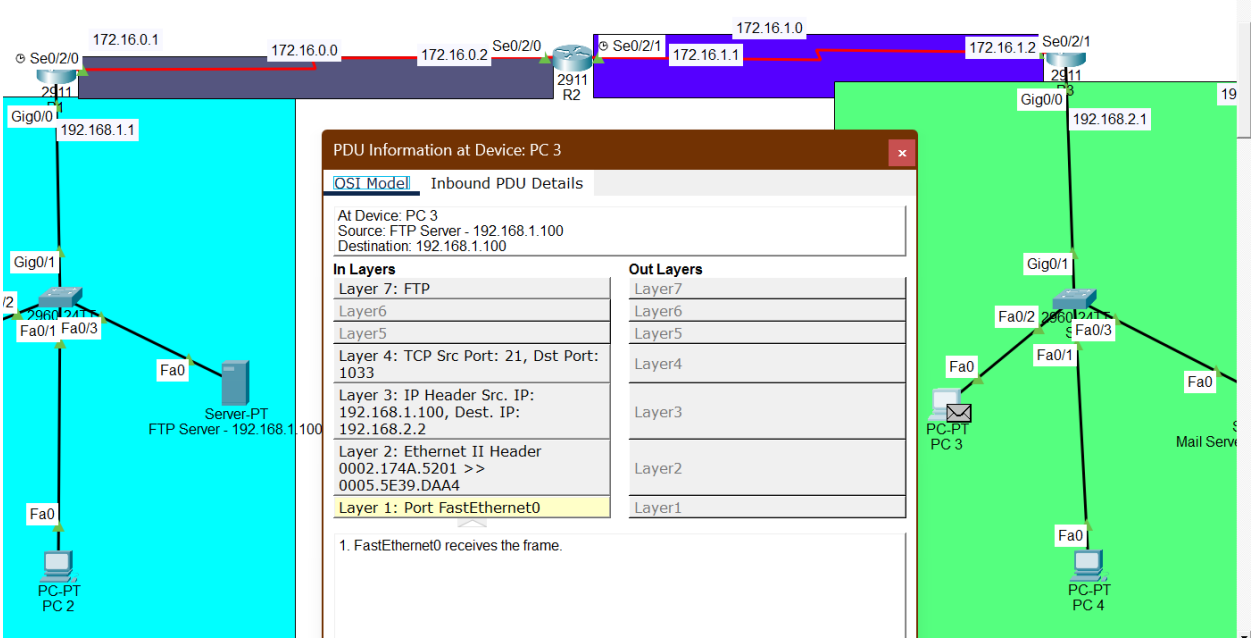

172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
2911
R2
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
Gig0/0
192.168.1.1
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1.100
PC-PT
PC 2
Gig0/0
192.168.2.1
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
Mail Serv
PC-PT
PC 4
PDU Information at Device: S1
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: S1
Source: FTP Server - 192.168.1.100
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Ethernet II Header
00D0.D395.B537 >>
00D0.58DB.1C01
Layer 1: Port FastEthernet0/3
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer 2: Ethernet II Header
00D0.D395.B537 >>
00D0.58DB.1C01
Layer 1: Port(s):
GigabitEthernet0/1
1. FastEthernet0/3 receives the frame.


172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
2911
R2
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
Gig0/0
192.168.1.1
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1.100
PC-PT
PC 2
Gig0/0
192.168.2.1
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
Mail Serv
PC-PT
PC 4
PDU Information at Device: R1
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: R1
Source: FTP Server - 192.168.1.100
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP:
192.168.1.100, Dest. IP:
192.168.2.2
Layer 2: Ethernet II Header
00D0.D395.B537 >>
00D0.58DB.1C01
Layer 1: Port GigabitEthernet0/0
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP:
192.168.1.100, Dest. IP:
192.168.2.2
Layer 2: HDLC Frame HDLC
Layer 1: Port(s): Serial0/2/0
1. GigabitEthernet0/0 receives the frame.


172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
Gig0/0
192.168.1.1
Gig0/0
192.168.2.1
PDU Information at Device: R2
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: R2
Source: FTP Server - 192.168.1.100
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.100, Dest. IP: 192.168.2.2
Layer 2: HDLC Frame HDLC
Layer 1: Port Serial0/2/0
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.100, Dest. IP: 192.168.2.2
Layer 2: HDLC Frame HDLC
Layer 1: Port(s): Serial0/2/1
1. Serial0/2/0 receives the frame.
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1.100
Fa0
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
Fa0
Fa0


172.16.0.1
Se0/2/0
172.16.0.0
172.16.0.2
Se0/2/0
Se0/2/1
172.16.1.0
172.16.1.1
172.16.1.2
Se0/2/1
Gig0/0
192.168.1.1
Gig0/0
192.168.2.1
PDU Information at Device: R3
OSI Model
Inbound PDU Details
Outbound PDU Details
At Device: R3
Source: FTP Server - 192.168.1.100
Destination: 192.168.1.100
In Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.100, Dest. IP: 192.168.2.2
Layer 2: HDLC Frame HDLC
Layer 1: Port Serial0/2/1
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.1.100, Dest. IP: 192.168.2.2
Layer 2: Ethernet II Header 0002.174A.5201 >> 0005.5E39.DAA4
Layer 1: Port(s): GigabitEthernet0/0
1. Serial0/2/1 receives the frame.
Gig0/1
Fa0/1
Fa0/3
Fa0
Server-PT
FTP Server - 192.168.1.100
Fa0
PC-PT
PC 2
Gig0/1
Fa0/2
Fa0/3
Fa0/1
Fa0
PC-PT
PC 3
Fa0
PC-PT
PC 4



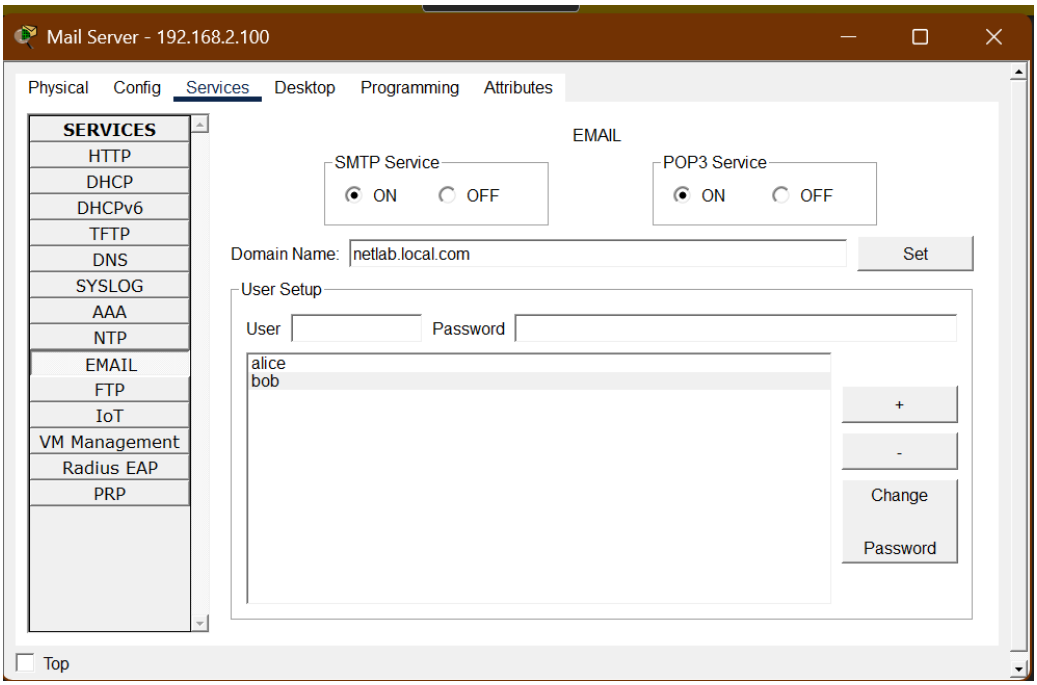


Now client has received the request for the username

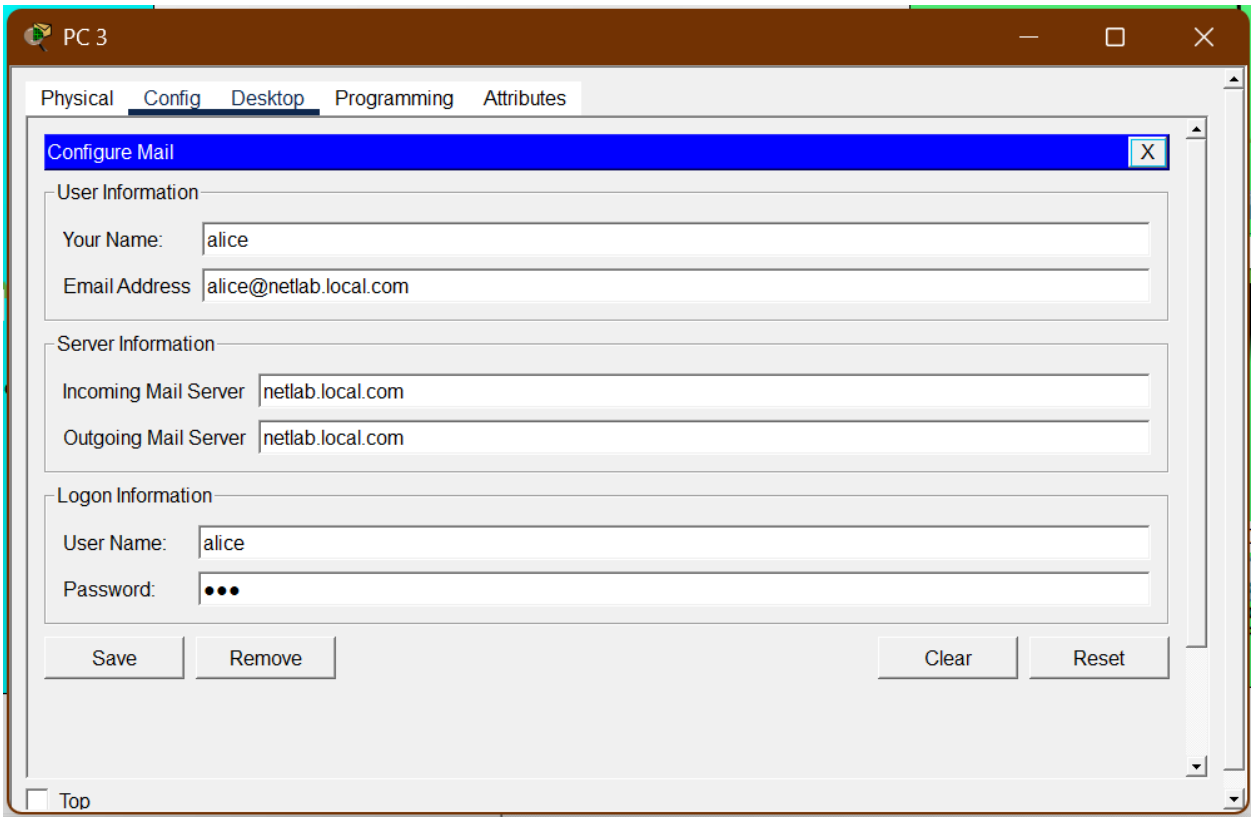
```
C:\>ftp ahsan-53143
Trying to connect...ahsan-53143
Connected to ahsan-53143
220- Welcome to PT Ftp server
Username:
```

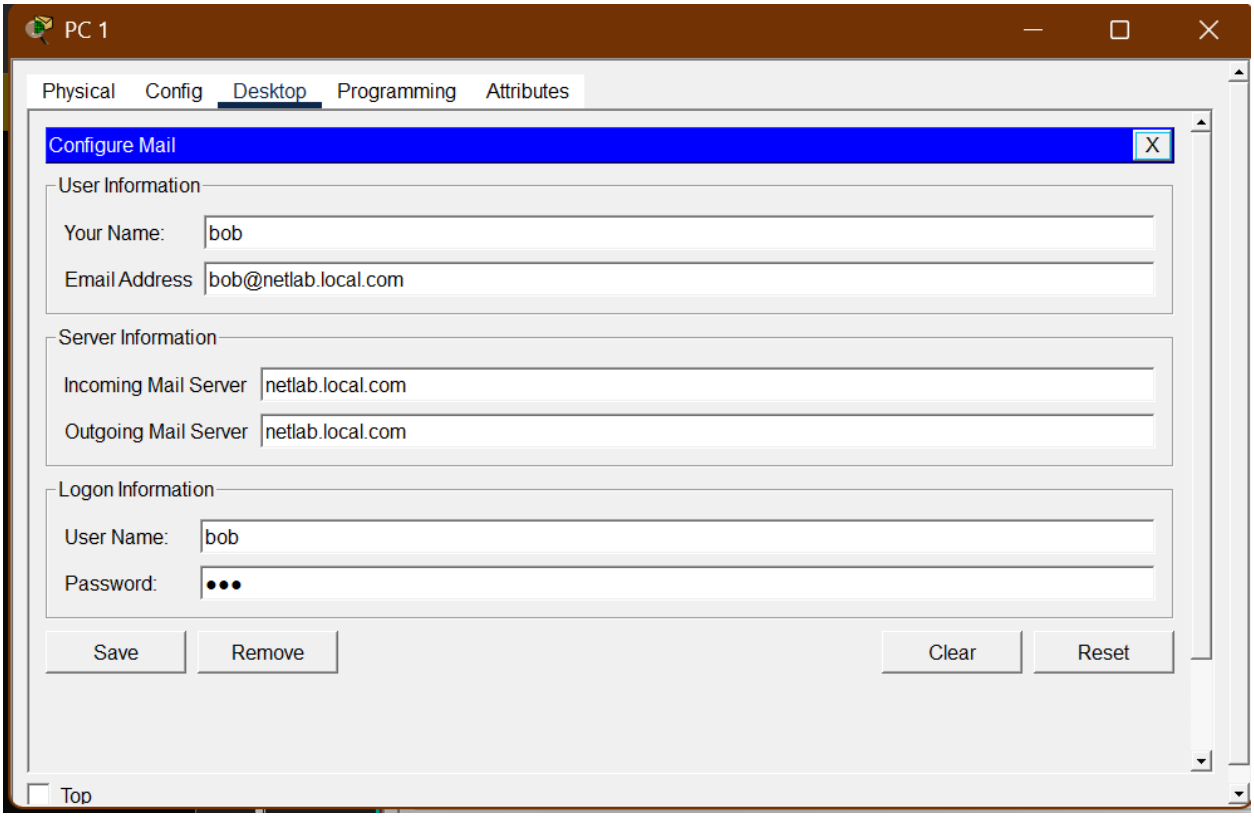
Task 3 — Mail Server & SMTP Configuration

1. On the Mail Server, set the domain name to netlab.local and enable both SMTP and POP3 services.
2. Create two mail accounts: alice@netlab.local and bob@netlab.local .

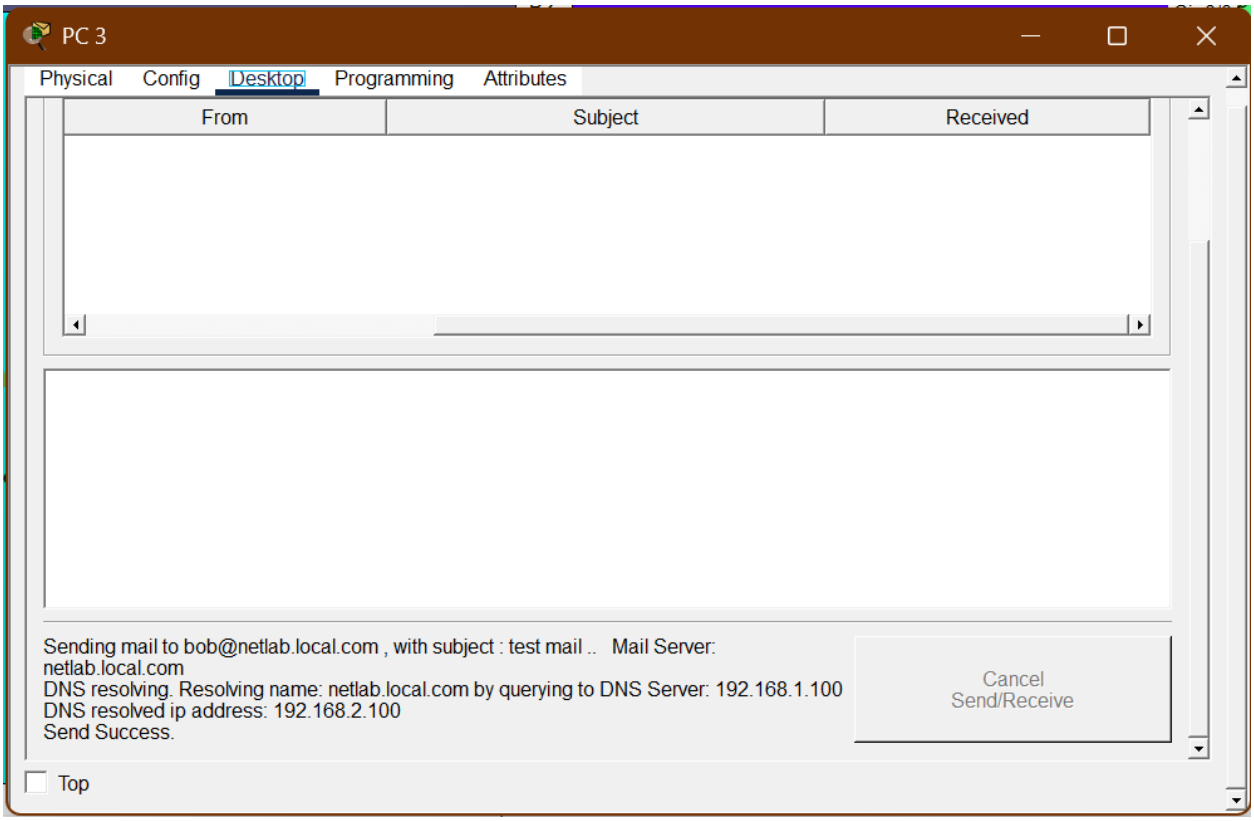


3. Configure the Mail client on PC-C for Alice and on PC-A for Bob, pointing both to the mail server for incoming and outgoing mail.

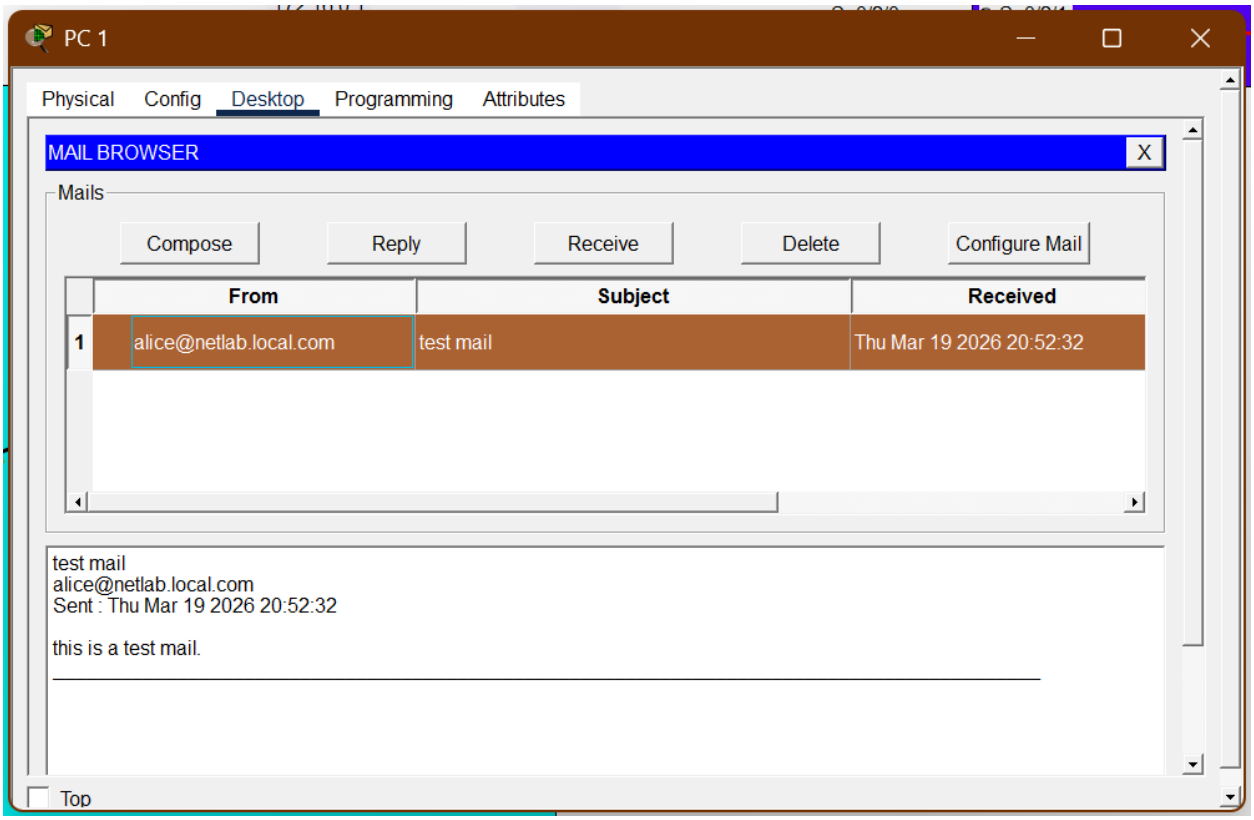




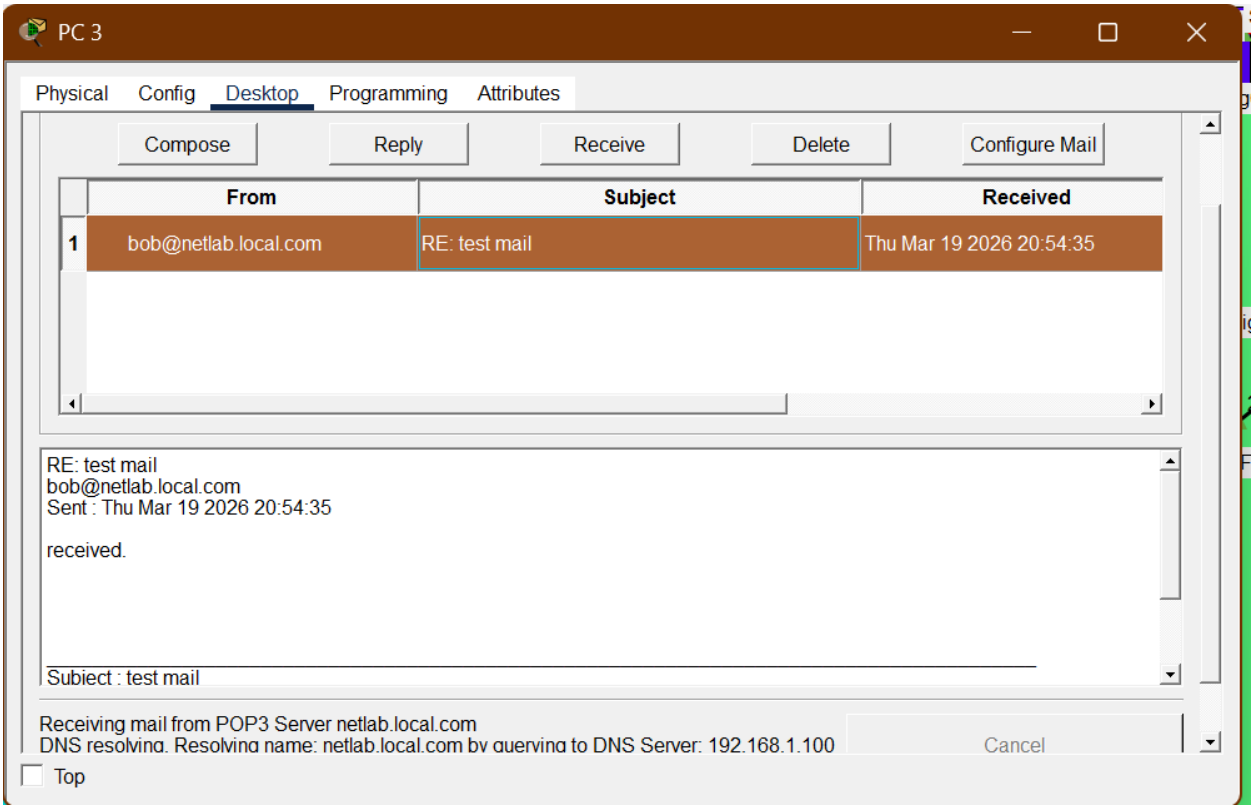
4. From Alice's client, send an email to bob@netlab.local .



5. From Bob's client, receive the email and send a reply to Alice.

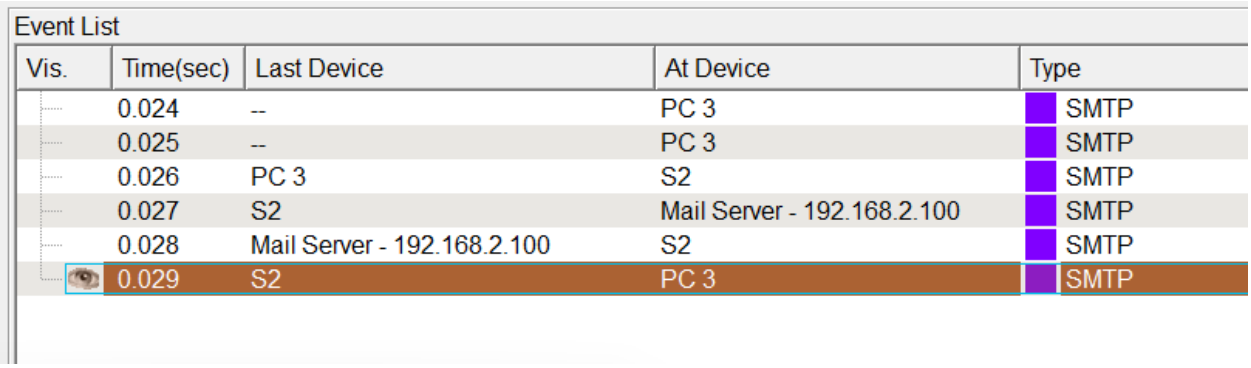


6. Confirm Alice receives Bob's reply on PC-C.



7. Using Simulation Mode, capture an SMTP PDU during message transmission and identify the destination port number. Explain in your report why that port is used.

SMTP Route:



Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.024	--	PC 3	SMTP
	0.025	--	PC 3	SMTP
	0.026	PC 3	S2	SMTP
	0.027	S2	Mail Server - 192.168.2.100	SMTP
	0.028	Mail Server - 192.168.2.100	S2	SMTP
	0.029	S2	PC 3	SMTP

POP3 Route:

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.016	--	PC 1	POP3
	0.017	--	PC 1	POP3
	0.018	PC 1	S1	POP3
	0.019	S1	R1	POP3
	0.020	R1	R2	POP3
	0.021	R2	R3	POP3
	0.022	R3	S2	POP3
	0.023	S2	Mail Server - 192.168.2.100	POP3
	0.024	Mail Server - 192.168.2.100	S2	POP3
	0.025	S2	R3	POP3
	0.026	R3	R2	POP3
	0.027	R2	R1	POP3
	0.028	R1	S1	POP3
	0.029	S1	PC 1	POP3

Reason for the port 21:

In networking, ports act like specific "service doors" on a building (the Server). Port 25 is the globally recognized door for SMTP (Simple Mail Transfer Protocol). When the PC sends data to Port 25, it is telling the server: "This information is a letter, please give it to the Mail Manager." If we used a different port, like Port 21 (which is for files), the Mail Server wouldn't understand the message because it would be expecting a file, not an email. Using Port 25 ensures that the mail is handled correctly and securely.

Task 4 — Verification & Fault Diagnosis

4.1 End-to-End Connectivity Verification

The following table documents the successful connectivity across the entire multi-service network using the standard ping tool from the Command Prompt of each end device.

Source Device	Destination Device	Destination IP	Result	Status
PC-A (LAN 1)	FTP Server	192.168.1.100	4/4 Received	**Pass**
PC-B (LAN 1)	Mail Server	192.168.2.100	4/4 Received	**Pass**
PC-C (LAN 2)	FTP Server	192.168.1.100	4/4 Received	**Pass**
PC-D (LAN 2)	Mail Server	192.168.2.100	4/4 Received	**Pass**

4.2 Written Fault Diagnosis

The following report details the identification and resolution of a "silent" network failure on the central router (R2).

1. Identification of the Issue:

- **Symptom**: Complete loss of connectivity between the R1 LAN segment (PC-A/B) and the R3 LAN segment (PC-C/D).
- **Testing**: Initiated a ping from PC-A to the Mail Server (192.168.2.100). The result was "Request Timed Out," indicating a break in the routed path.

2. Troubleshooting Steps & Findings:

- **Tool Used**: Cisco IOS command show ip interface brief on the central router (R2).
- **Diagnosis**: The command output revealed that the Serial interface connecting toward the R3 network (Serial 0/3/1) was in an **"Administratively Down"** status.
- **Routing Impact**: By using show ip route on R1, it was confirmed that the RIP-learned route for the 192.168.2.0/24 subnet had vanished because R2 was no longer able to send or receive routing updates through the disabled interface.

3. Corrective Action Taken:

- **Resolution**: I accessed the Global Configuration mode on Router R2 and entered the specific interface configuration.
- **Commands Executed**:

```
R2(config)# interface Serial0/3/1
R2(config-if)# no shutdown
```

- **Result**: The interface status changed to "Up/Up." Within a few moments, the RIP protocol reconverged, the routing tables were updated, and end-to-end ping connectivity was successfully restored.

--- The End ---